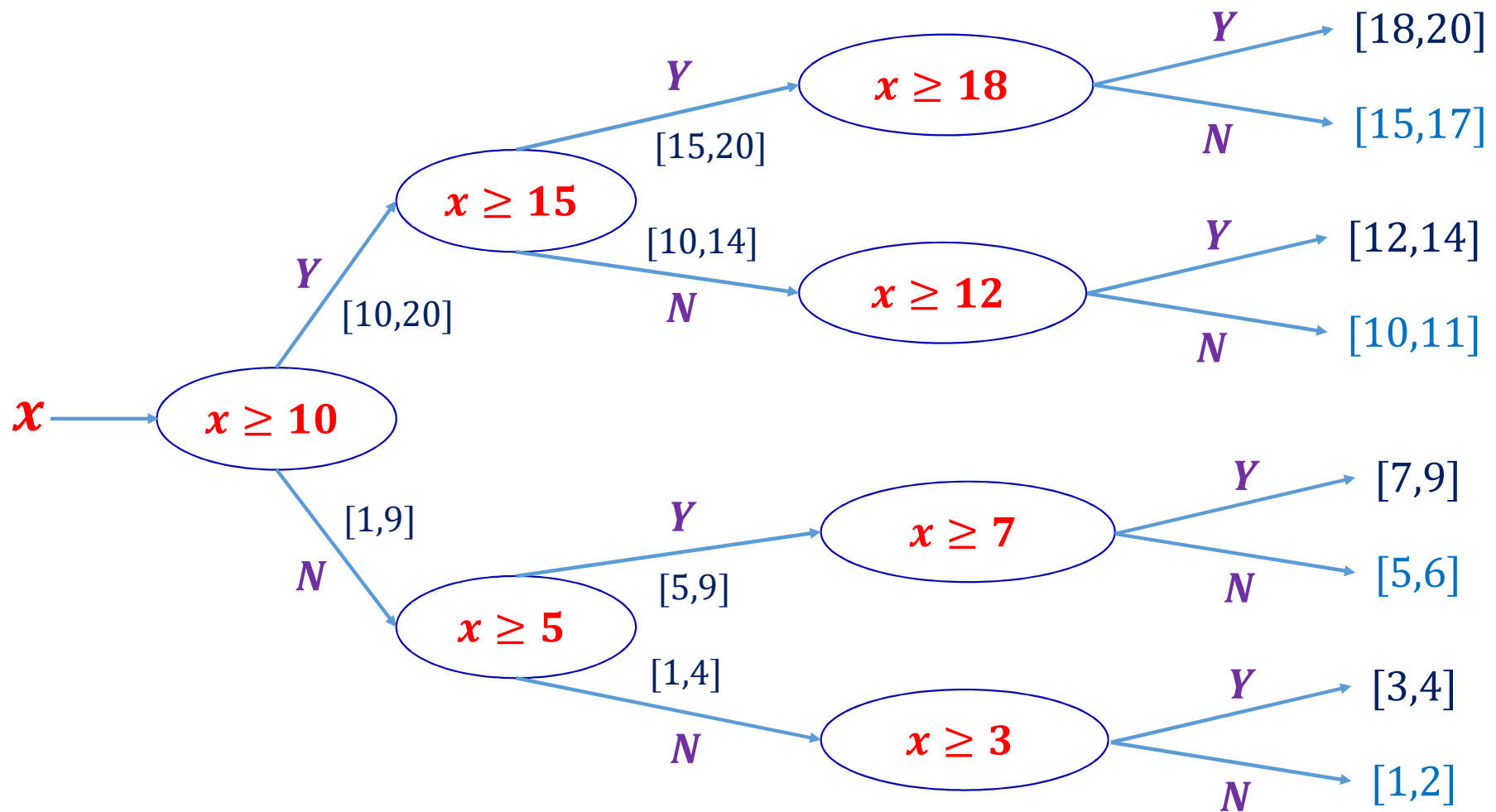


Classification & Regression Trees

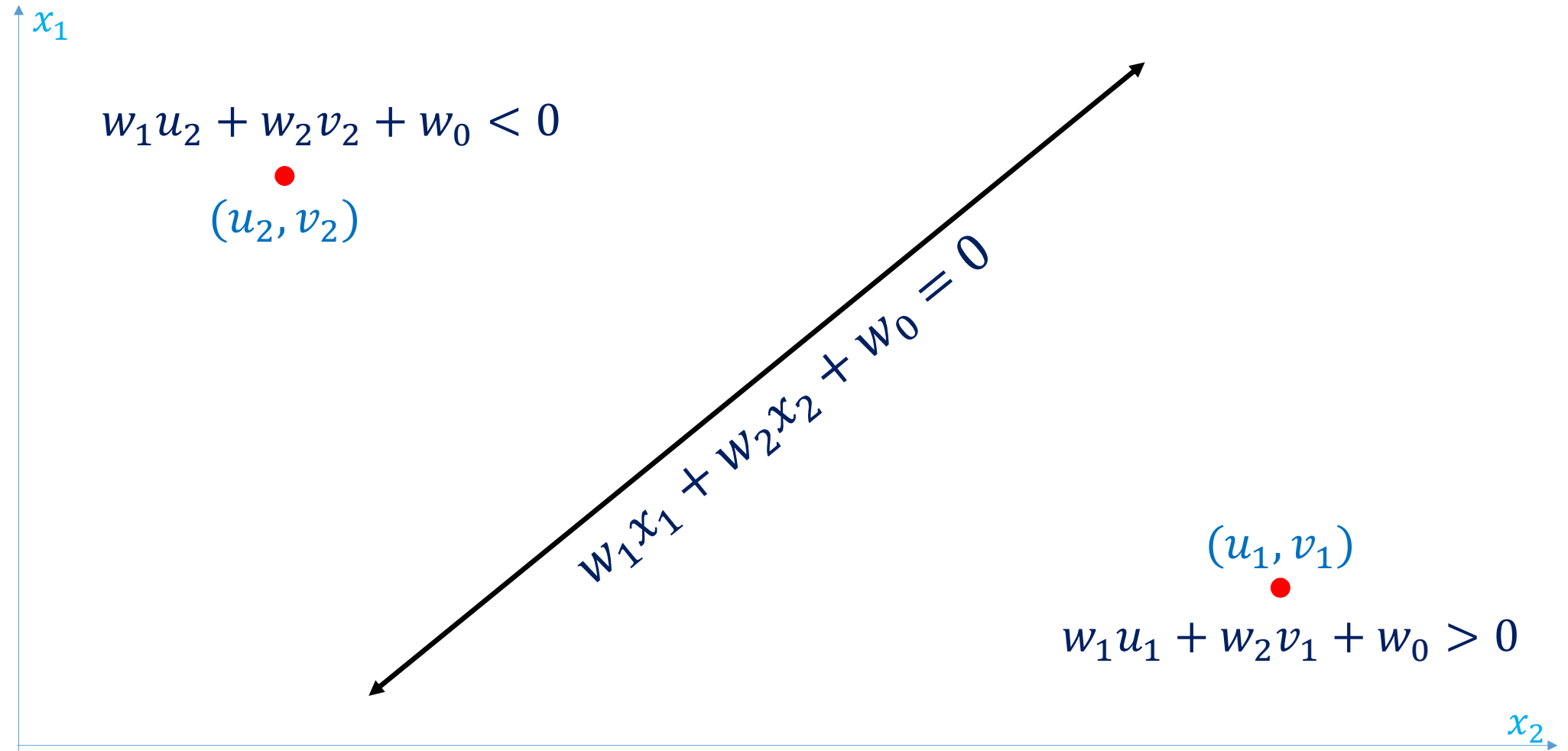


Prithwijit Guha
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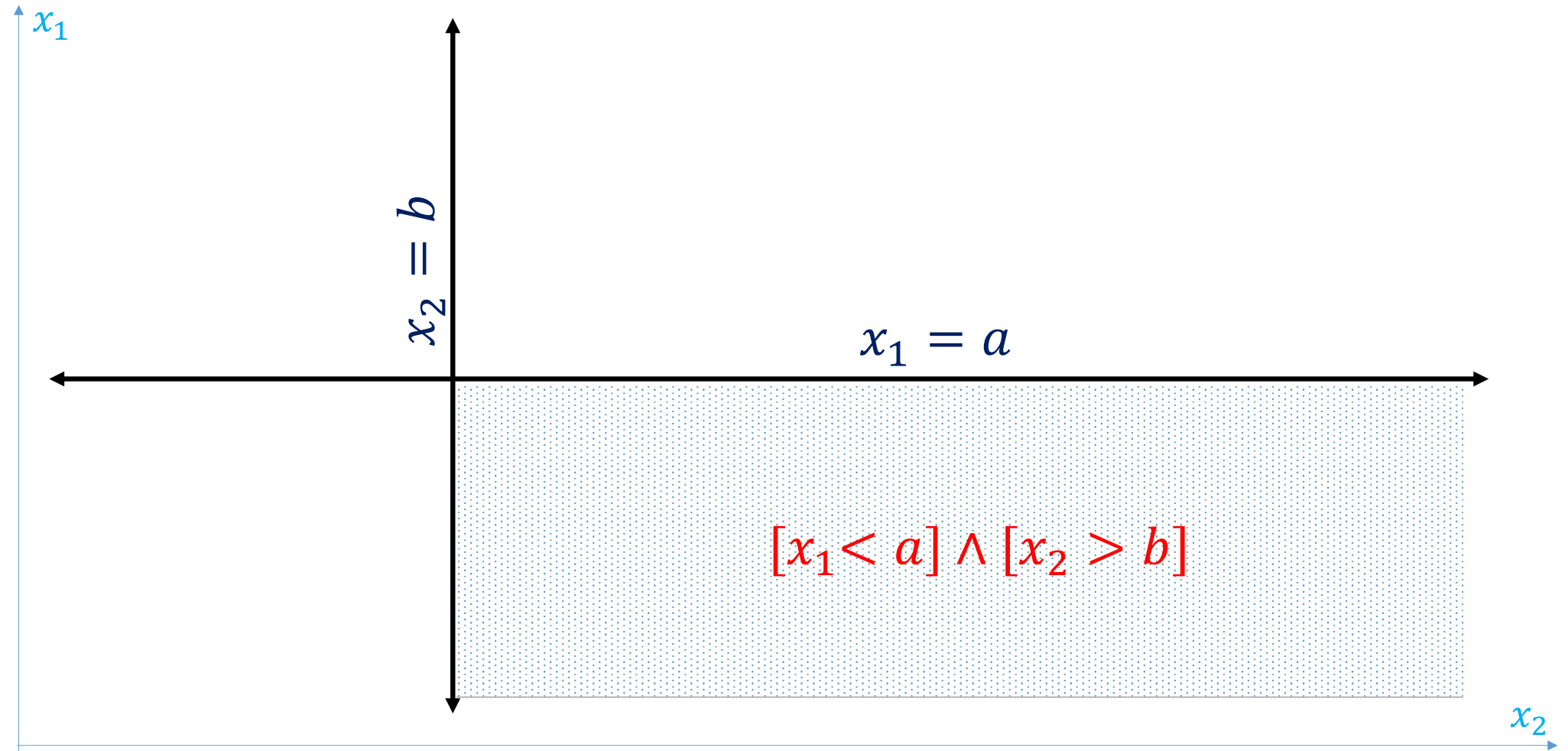
Search Space Partitioning



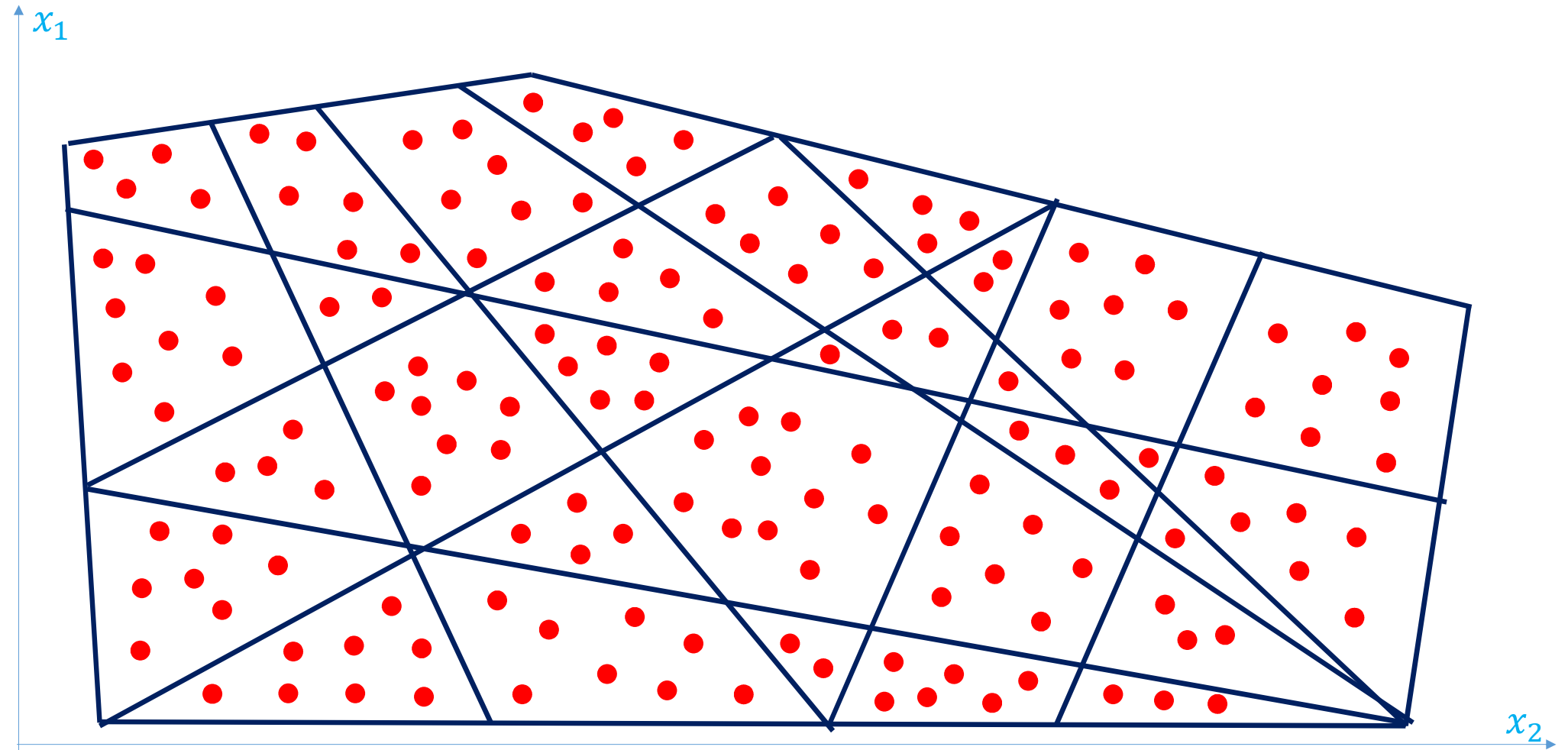
Feature Space Partitioning: Oblique



Feature Space Partitioning: Axis Aligned

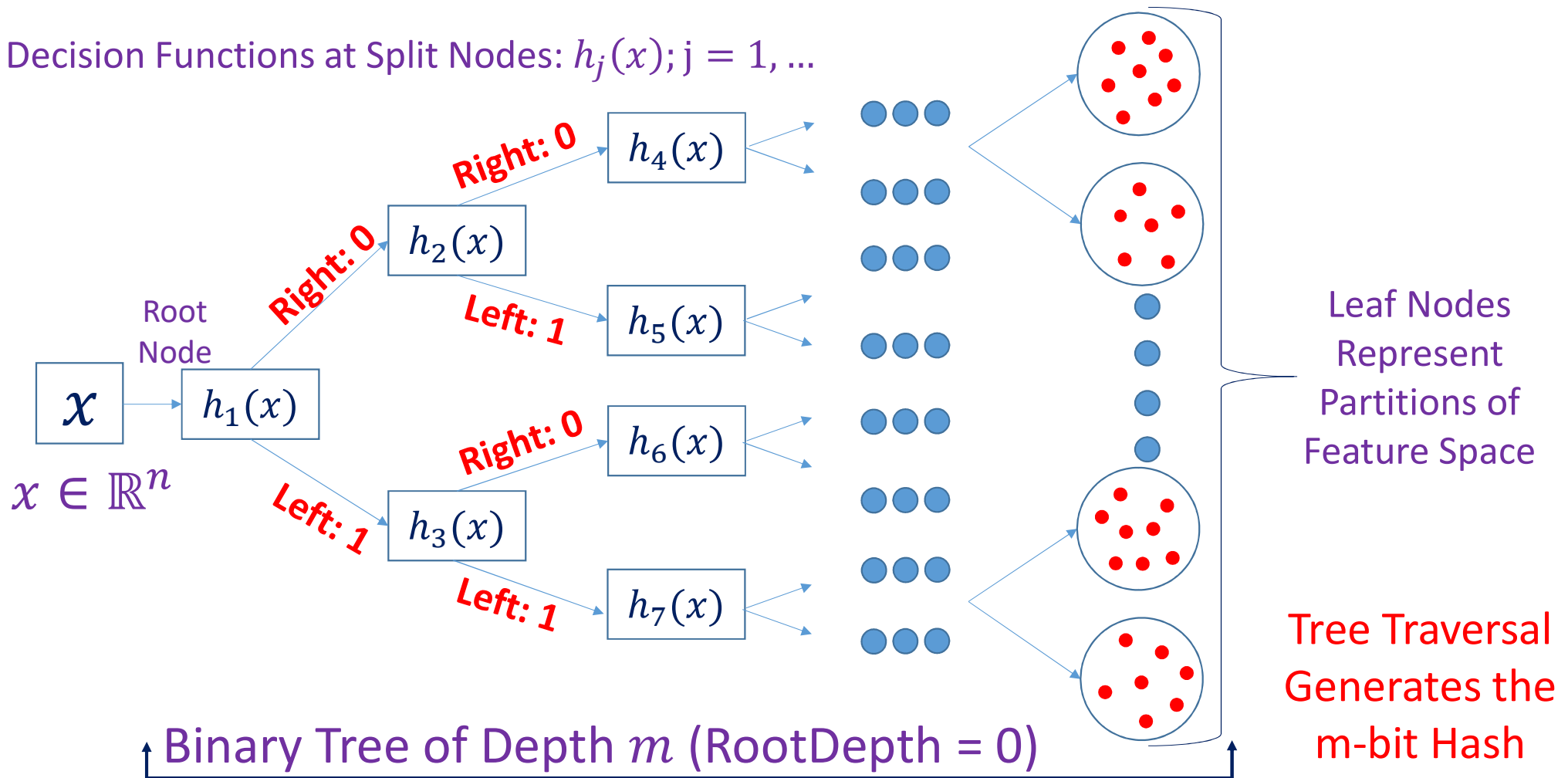


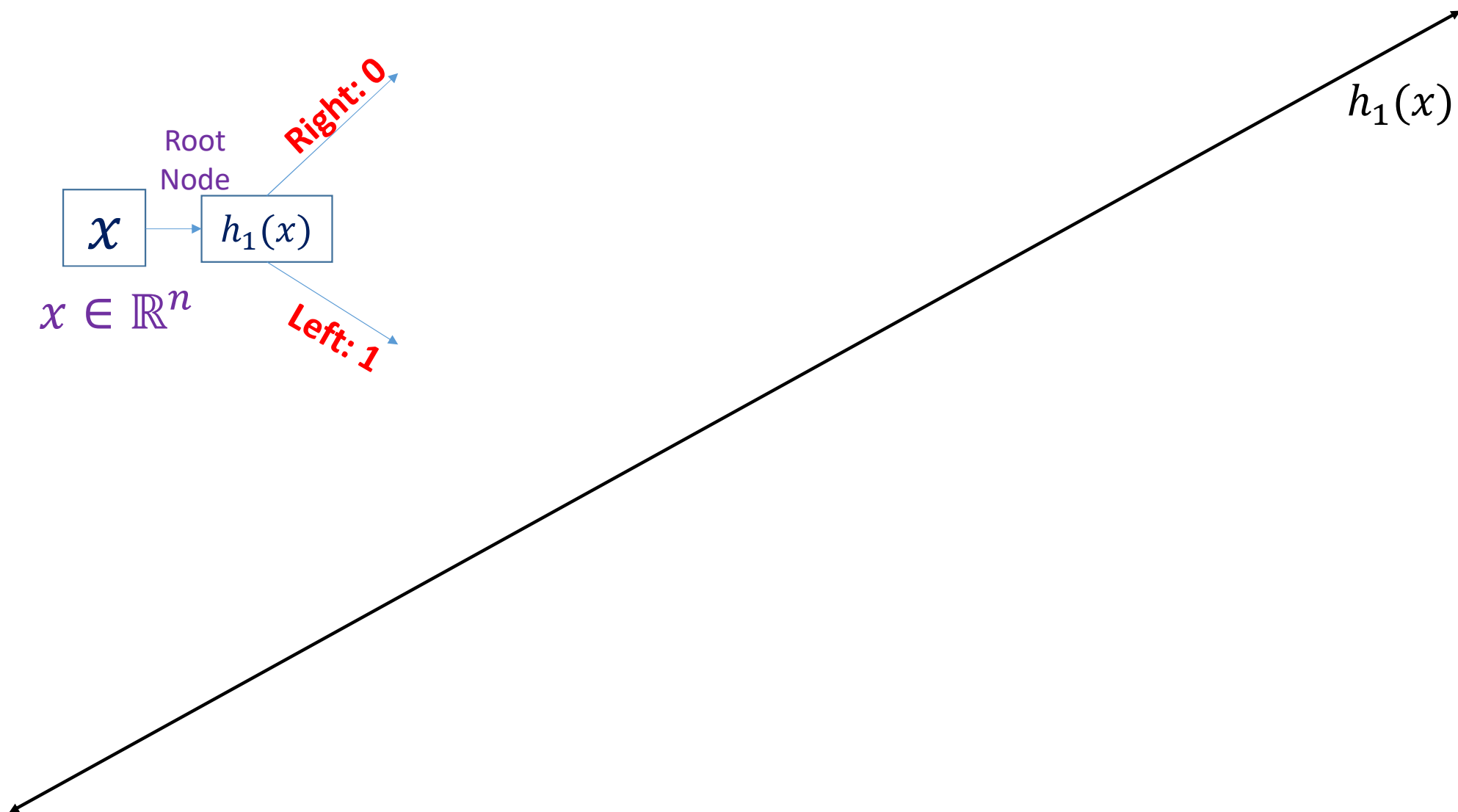
Feature Space Partitioning

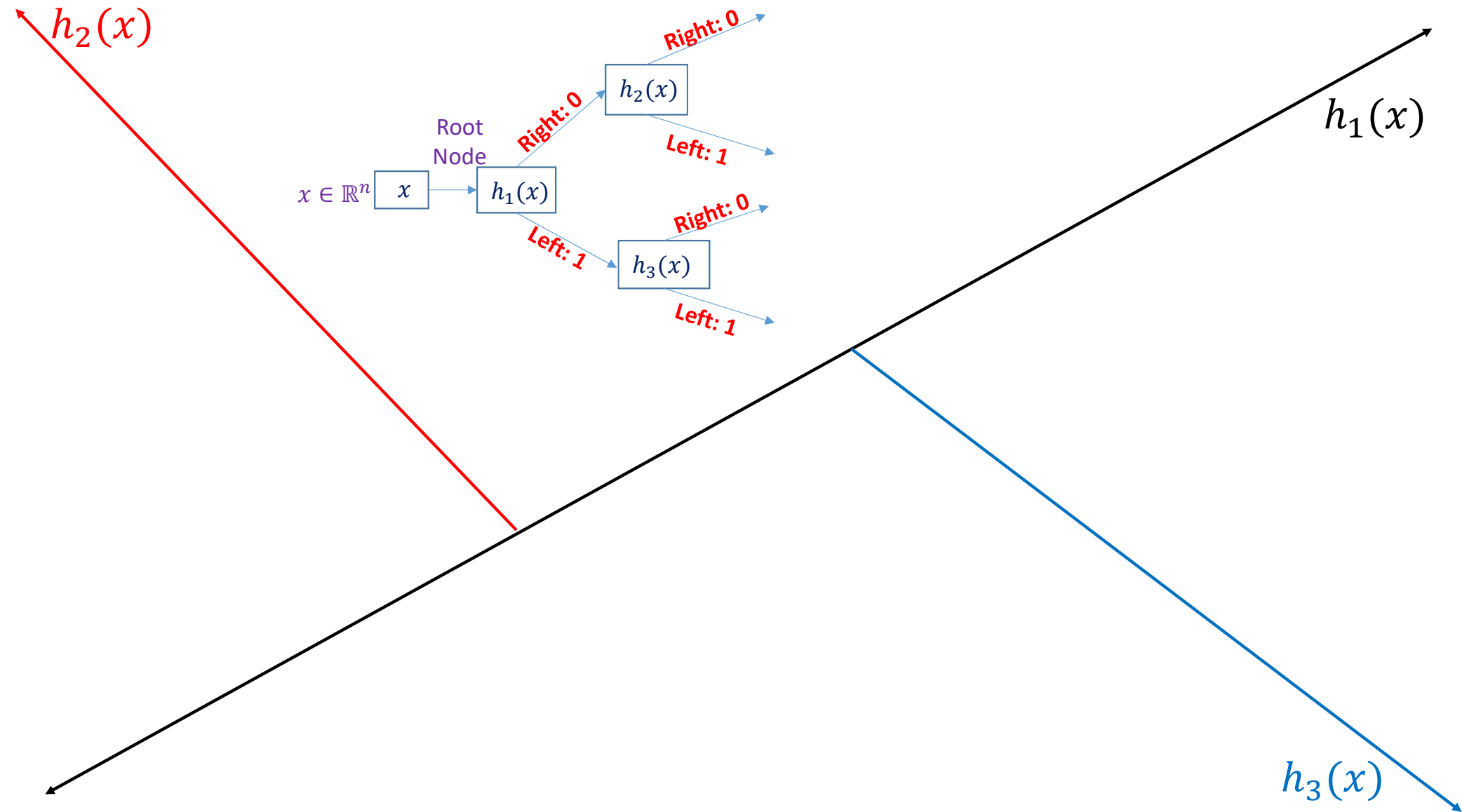


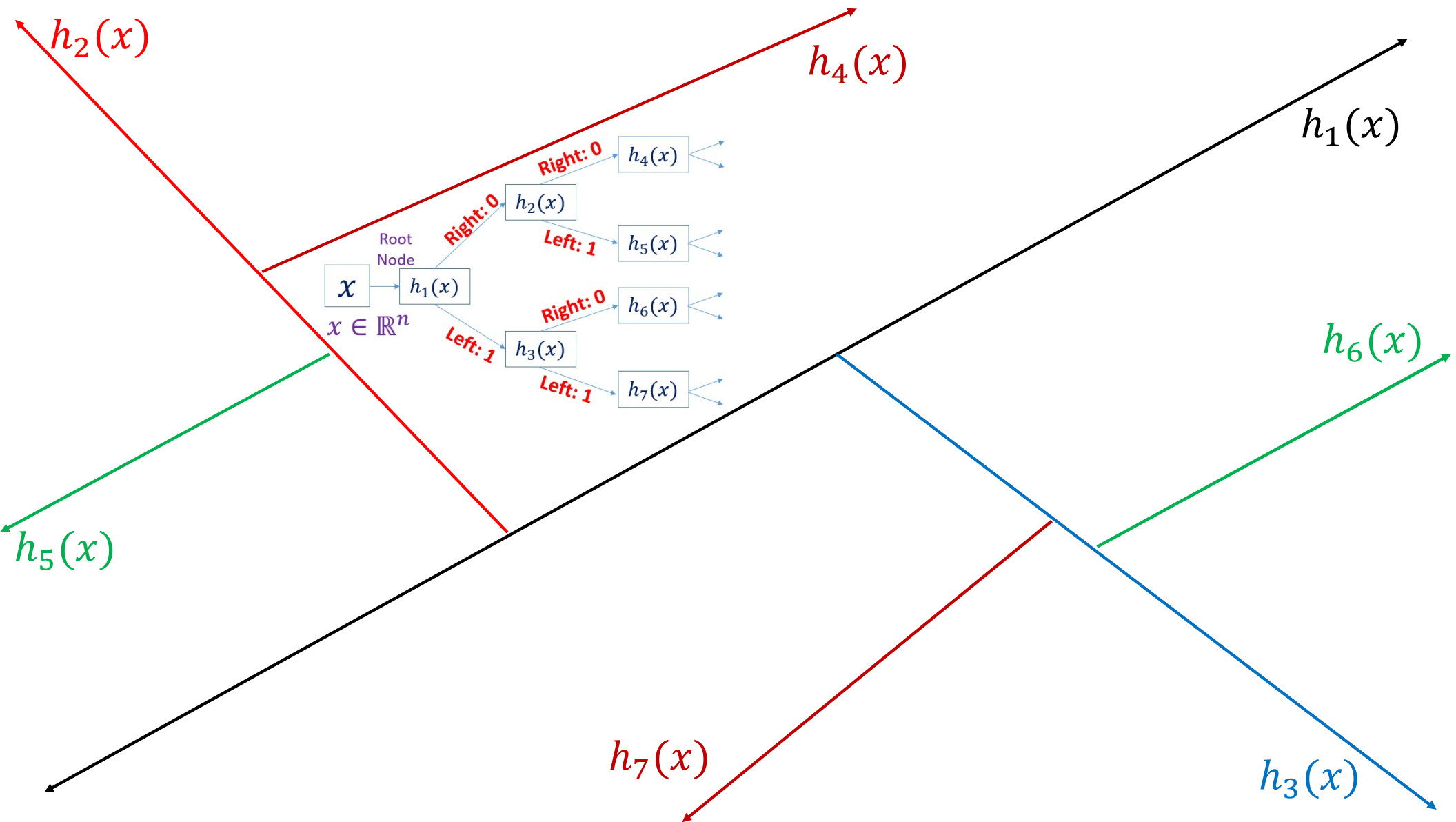
Top-Down Hierarchical Partitioning of Feature Space

Decision Functions at Split Nodes: $h_j(x); j = 1, \dots$







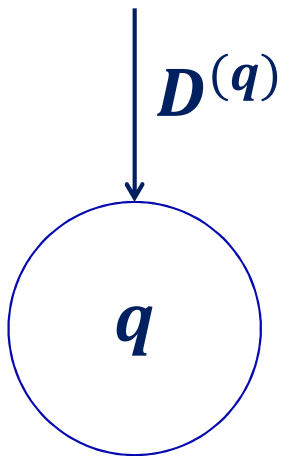


KD-Tree Algorithm: Termination Criterion

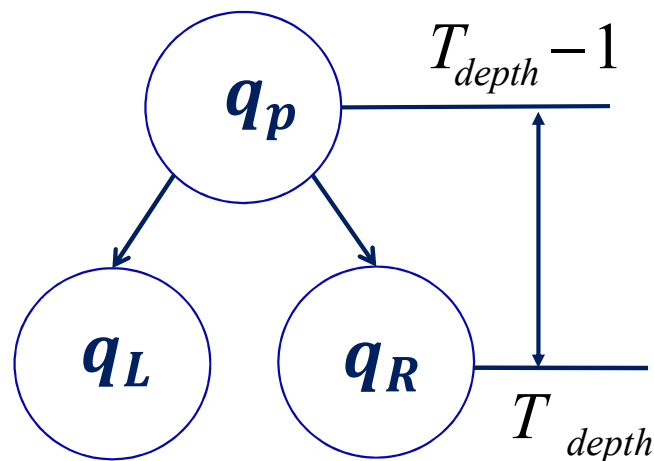
Maximum Tree Depth is T_{depth}^{max}

Input to Node q of Tree T is $D^{(q)} = \{x_i; i = 1, \dots, n_q; x \in \mathbb{R}^d\}$

Minimum Number of Data Instance Necessary for a Split is m



Insufficient Input to
Node $|D^{(q)}| < m$



Nodes at Maximum
Allowable Tree Depth

```
FUNCTION bool  $snc = splitNode(q)$   
IF  $[D^{(q)} < m] \vee [Depth(q) = T_{depth}^{max}]$   
    RETURN FALSE  
ELSE  
    RETURN TRUE  
ENDIF
```

KD-Tree Algorithm

INPUT: $\mathbf{D}^{(q)} = \{\mathbf{x}_i; i = 1, \dots, n_q, \mathbf{x} \in \mathbb{R}^d\}$

Evaluate Split Criteria: $snc = \text{splitNode}(\mathbf{q})$

IF $snc = \text{TRUE}$

Get Dimension of Highest Variance: $j_q = \underset{r=1 \dots d}{\operatorname{argmax}} \operatorname{Var} \{ \mathbf{x}_1[r], \dots, \mathbf{x}_{n_q}[r] \}$

Compute Median: $\theta_q = \operatorname{Median} \{ \mathbf{x}_1[j_q], \dots, \mathbf{x}_{n_q}[j_q] \}$

Construct Decision Function: $h_q(\mathbf{x}): \mathbf{x}[j_q] \geq \theta_q$

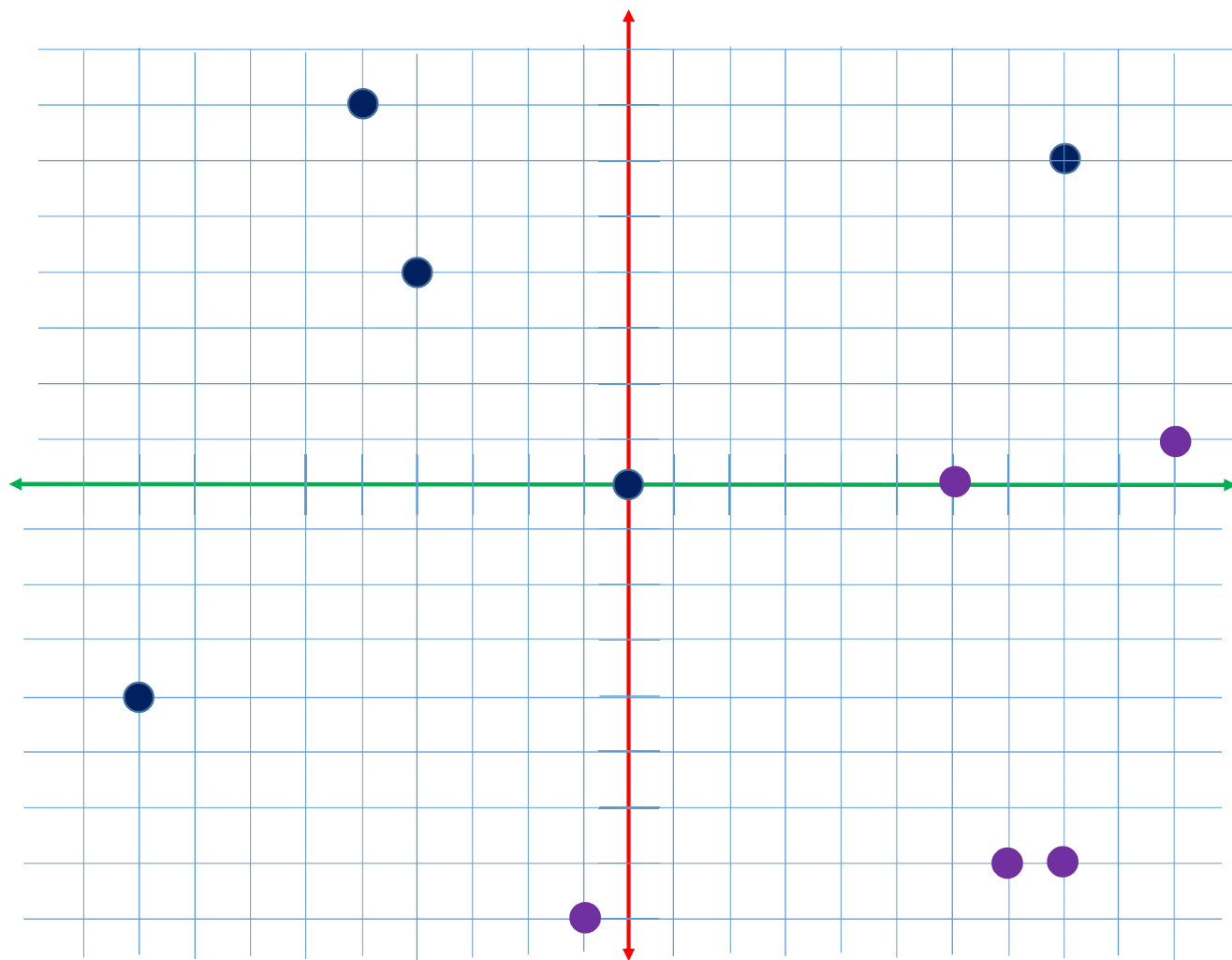
IF $\mathbf{x}[j_q] \geq \theta_q$ APPEND $\mathbf{x} \rightarrow \mathbf{D}_R^{(q)}$; ELSE APPEND $\mathbf{x} \rightarrow \mathbf{D}_L^{(q)}$

ENDIF

INPUT: $\mathbf{D}_L^{(q)} \rightarrow \mathbf{q}_L$ AND $\mathbf{D}_R^{(q)} \rightarrow \mathbf{q}_R$

The Dataset

6	-4	4	7	0	-7	1	-8	0	-7
8	-9	-4	-5	0	8	10	-1	6	7
+1	+1	+1	+1	+1	-1	-1	-1	-1	-1



6	-4	4	7	0	-7	1	-8	0	-7
8	-9	-4	-5	0	8	10	-1	6	7
+1	+1	+1	+1	+1	-1	-1	-1	-1	-1

The Dataset: Finding Split Parameters

6 -4 4 7 0 -7 1 -8 0 -7

8 -9 -4 -5 0 8 10 -1 6 7

+1 +1 +1 +1 +1 -1 -1 -1 -1 -1

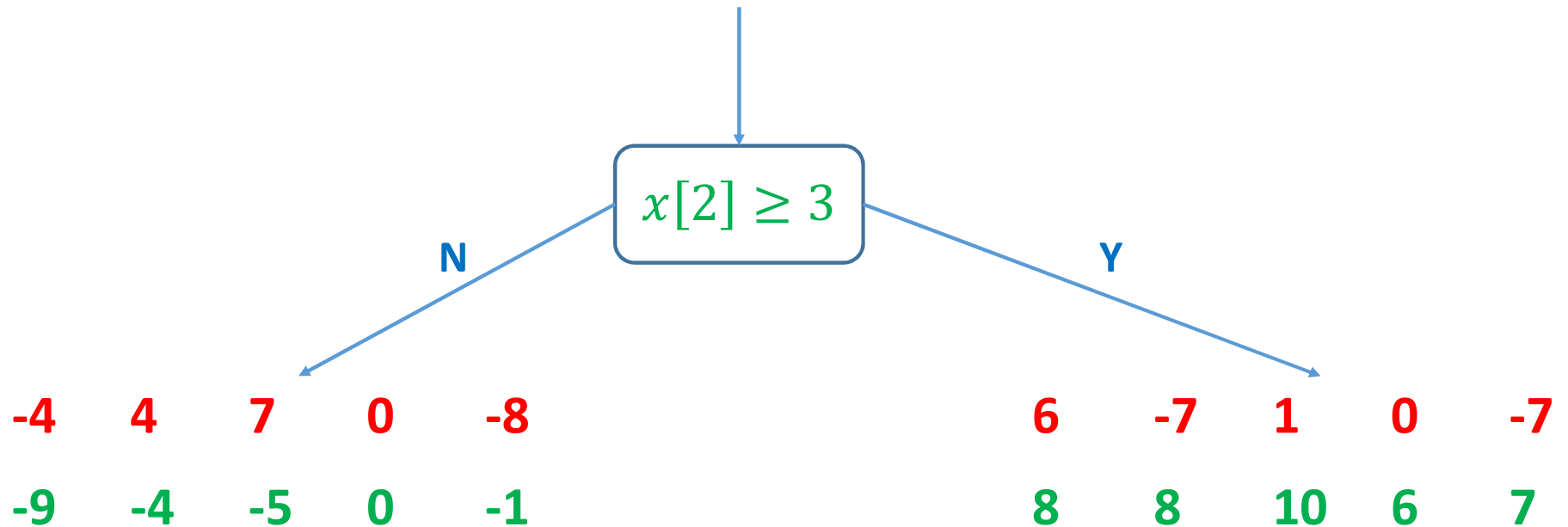
$$d = 1 \quad \sigma_1 = 5.5136$$

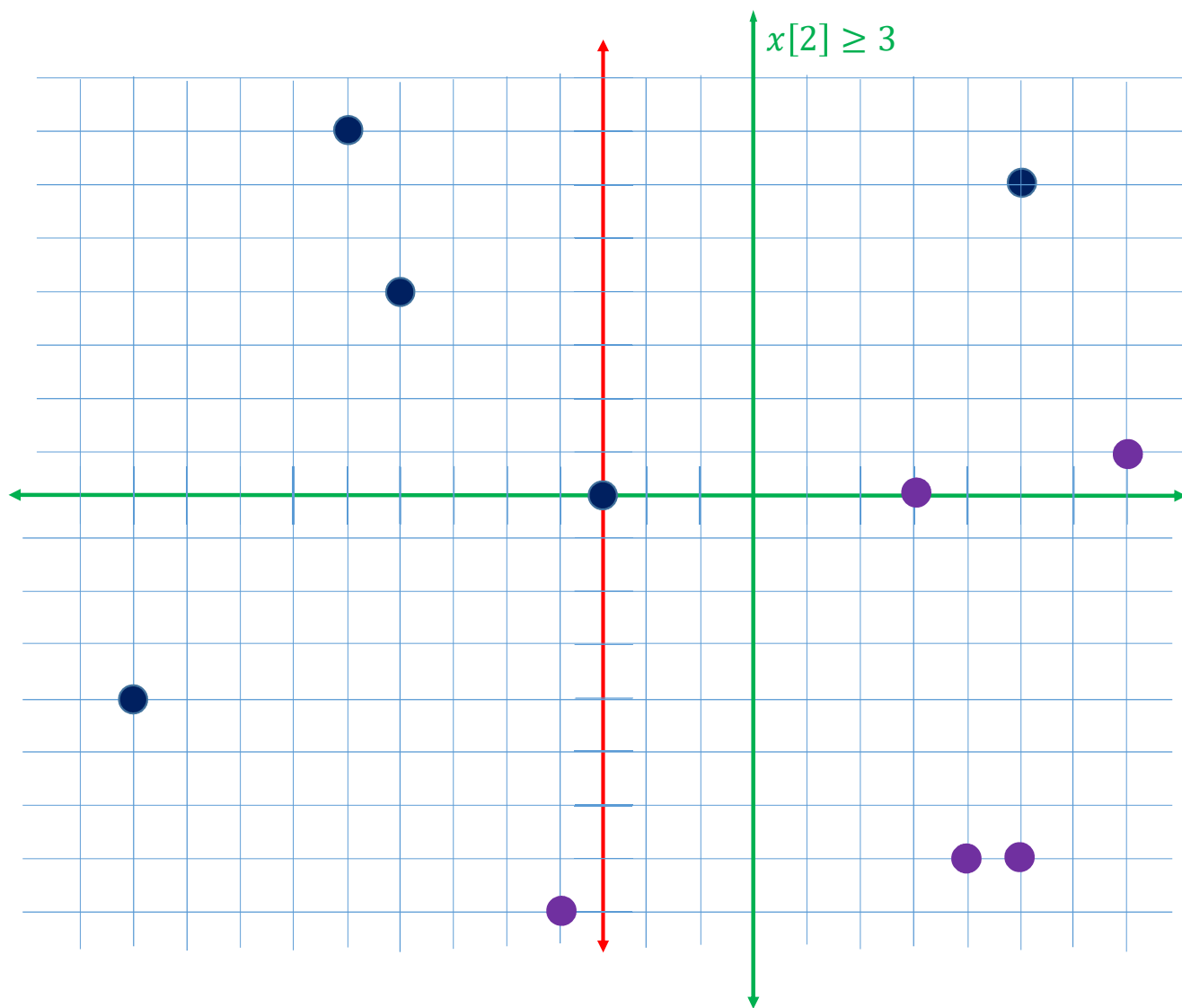
$$d = 2 \quad \sigma_2 = 6.6332$$

$$\text{median}[d = 2] = 3$$

KD-Tree: Data Splitting at Root Node

6	-4	4	7	0	-7	1	-8	0	-7
8	-9	-4	-5	0	8	10	-1	6	7





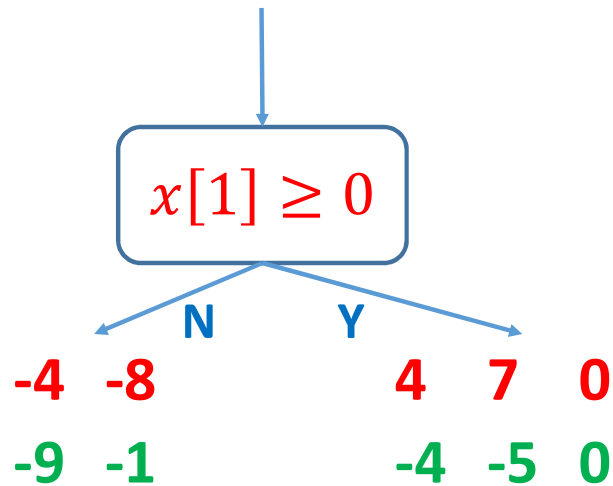
6	-4	4	7	0	-7	1	-8	0	-7
8	-9	-4	-5	0	8	10	-1	6	7
+1	+1	+1	+1	+1	-1	-1	-1	-1	-1

KD-Tree: Further Splitting

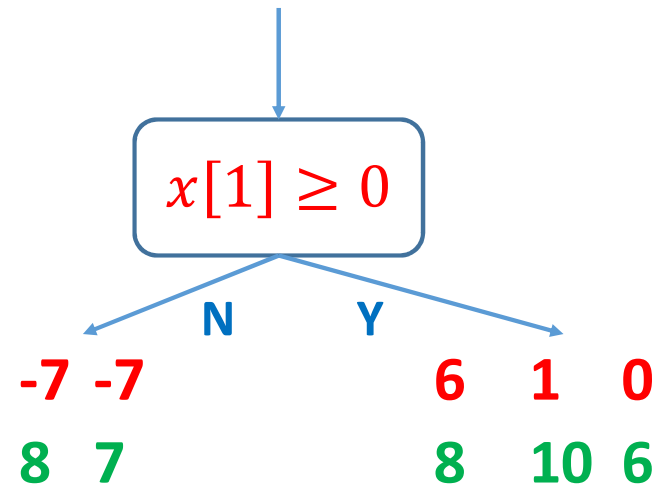
-4 4 7 0 -8 $\sigma_1 = 6.0166$
-9 -4 -5 0 -1 $\sigma_2 = 3.5637$

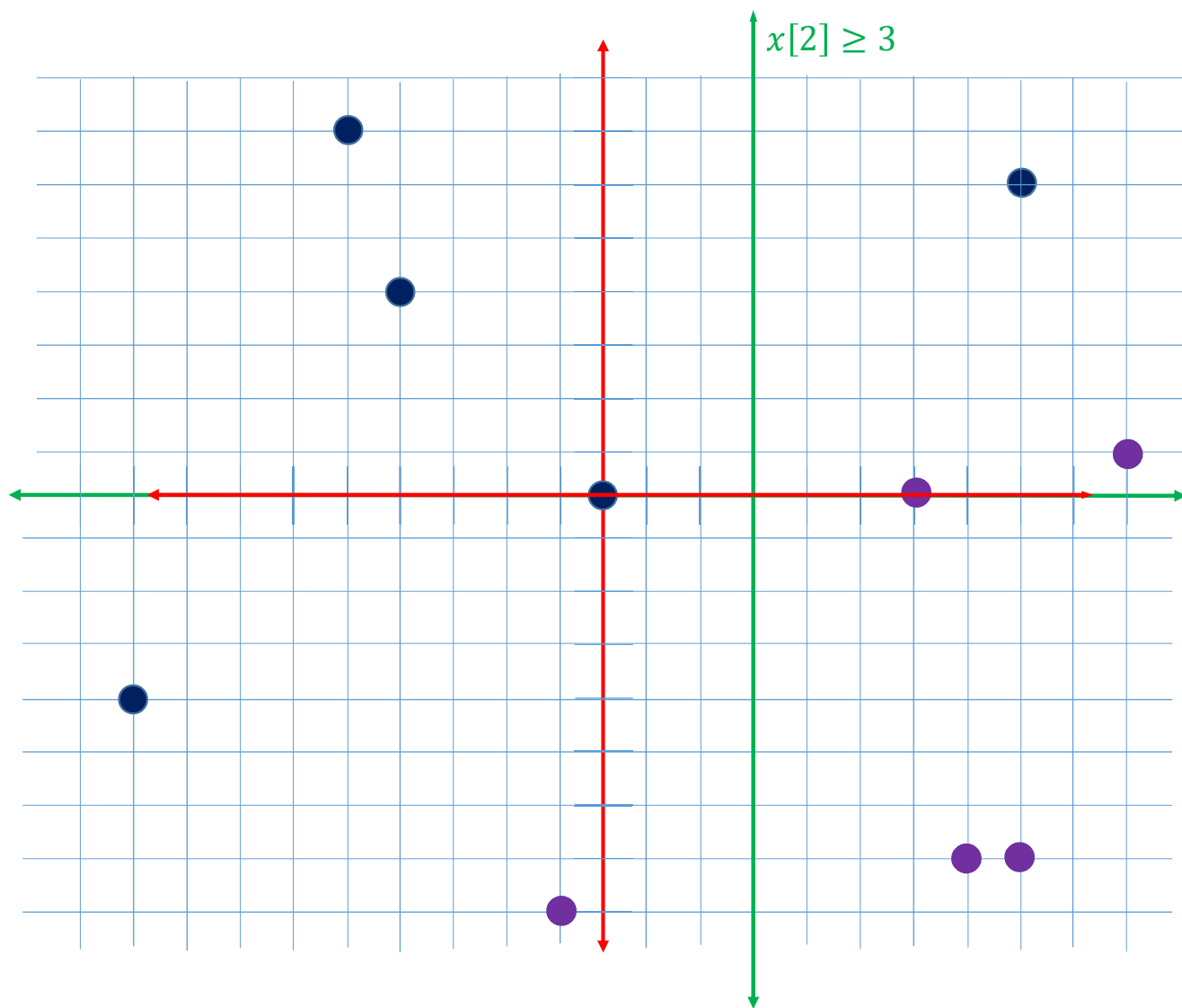
6 -7 1 0 -7 $\sigma_1 = 5.5946$
8 8 10 6 7 $\sigma_2 = 1.4832$

$median[d = 1] = 0$



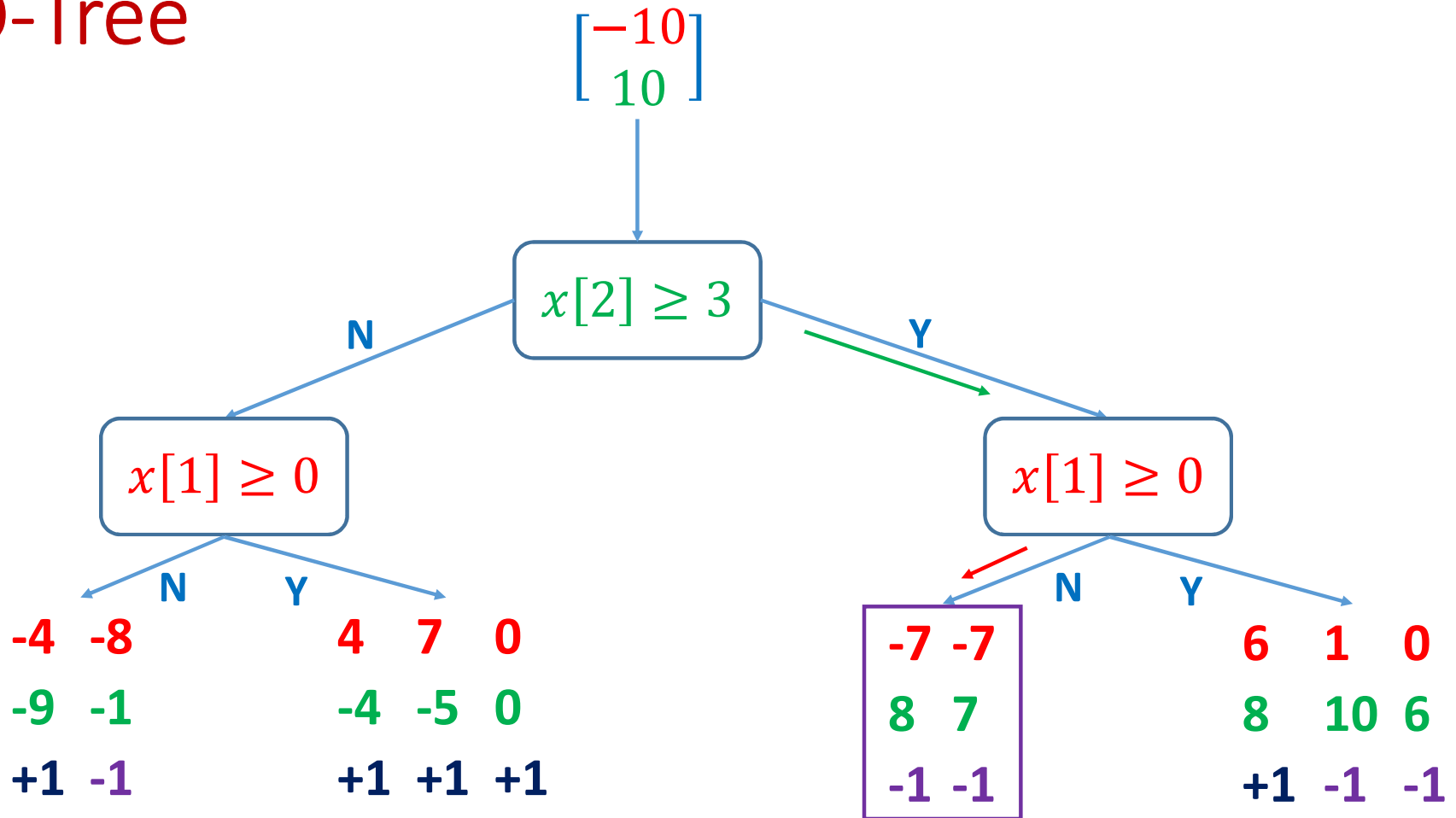
$median[d = 1] = 0$



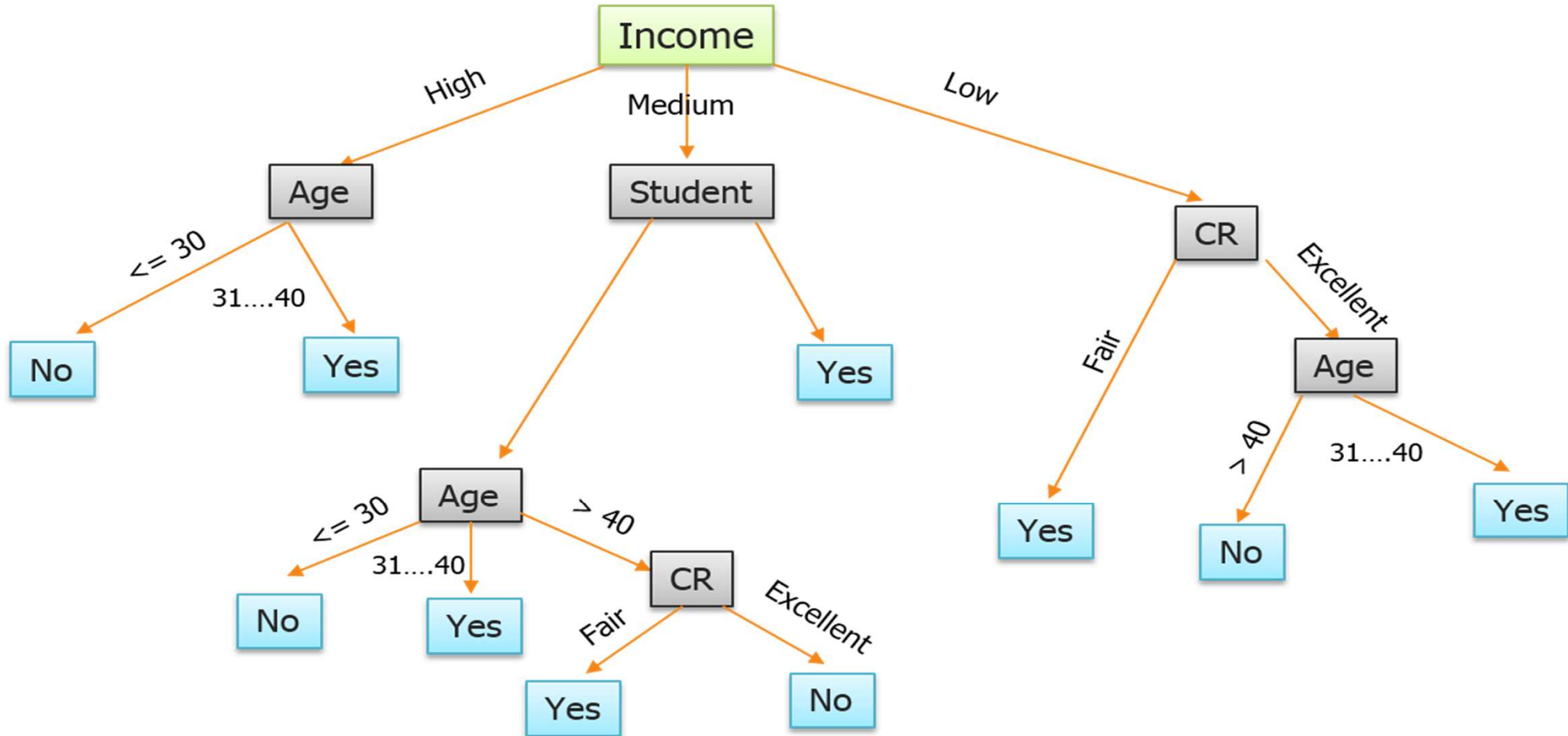


6	-4	4	7	0	-7	1	-8	0	-7
8	-9	-4	-5	0	8	10	-1	6	7
+1	+1	+1	+1	+1	-1	-1	-1	-1	-1

KD-Tree

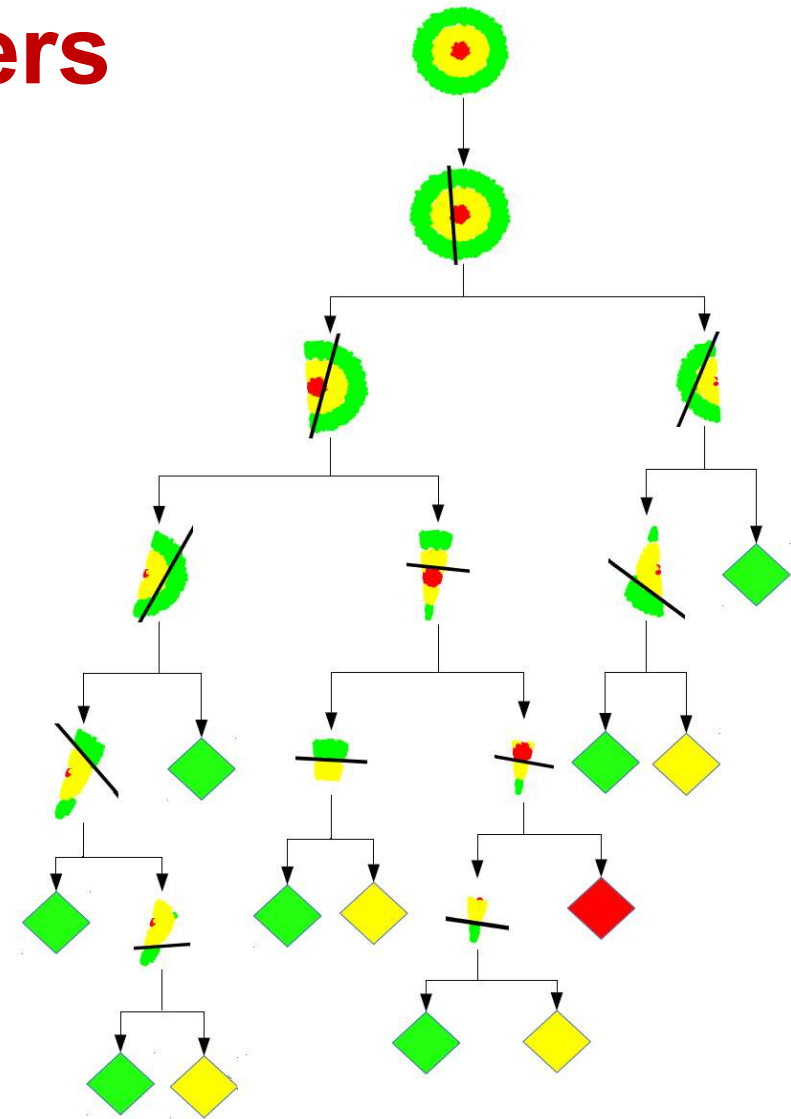
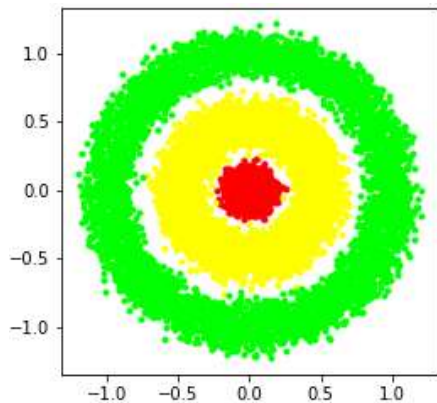


Decision Structures in Daily Life

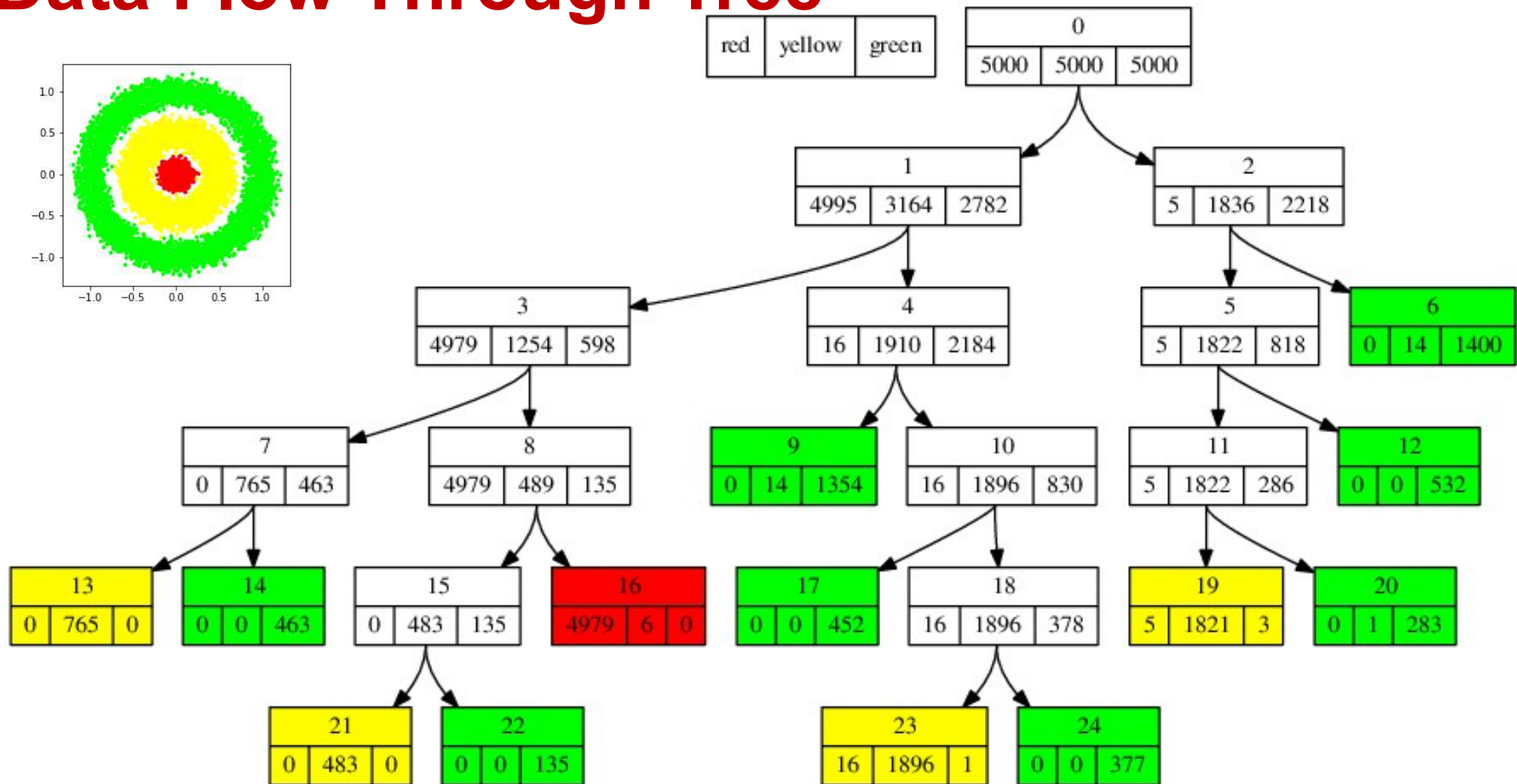
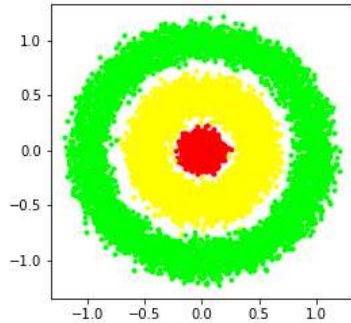


Decision Trees as Classifiers

- Hierarchical Top-Down Partitioning
- Decision Functions at Splitting Nodes
- Label Impurity Reduction in Partitions
- Leaf Nodes: “Pure” Partitions
- C4.5 Trees: Only Axis Aligned Splits
- Neural Trees: Allow Oblique Splits

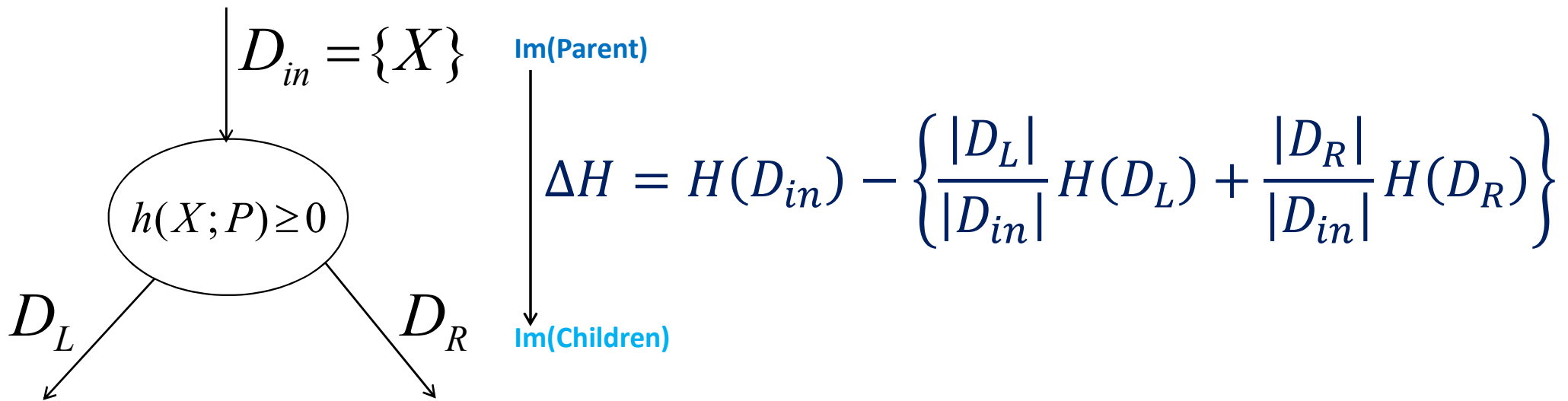


Data Flow Through Tree



Decision Tree: Impurity

$$D_{in} = \{(\mathbf{x}_i, y_i); i = 1, \dots | D_{in} |\}$$



Split Objective: Maximize Impurity Drop

Impurity Measures

Misclassification Impurity

$$H_c(D) = 1 - \max_{i=1,\dots,n} p_i$$

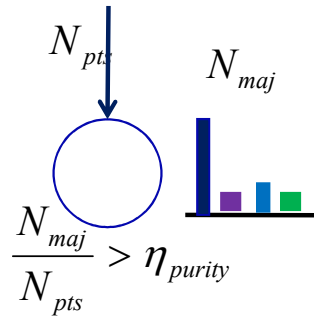
Gini Impurity

$$H_g(D) = 1 - \sum_{i=1}^n p_i^2$$

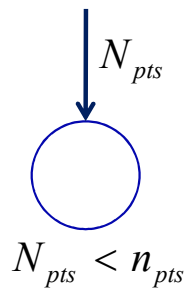
Entropy Impurity

$$H_e(D) = - \sum_{i=1}^n p_i \log_2 p_i$$

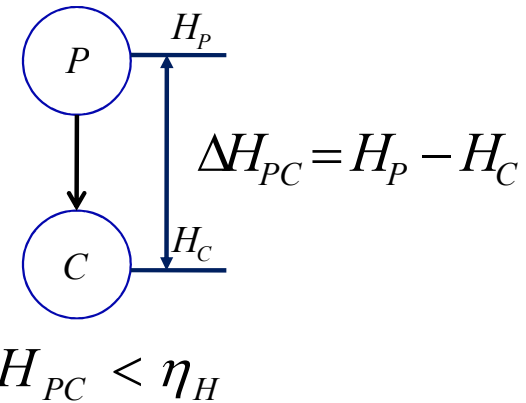
Termination Criteria



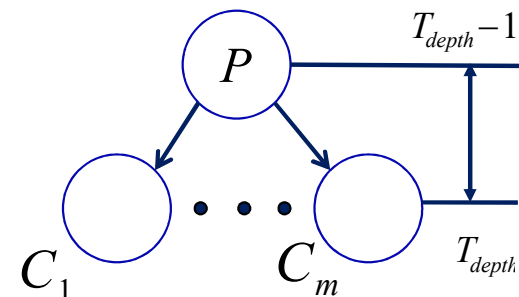
**Majority Class Exceeds
Purity Threshold**



**Insufficient Input
to Node**



**Insufficient Impurity Drop
from Parent to Child Node**



**Nodes at Maximum
Allowable Tree Depth**

Axis Aligned Decision Tree Construction

INPUT: $\mathbf{D}^{(q)}$ To Node q

IF $checkTerminationConditions(q) = TRUE$ THEN Tag q as Leaf Node

IF $isLeafNode(q) = TRUE$ THEN Tag q with Dominant Class Label of $\mathbf{D}^{(q)}$

ELSE Search for Optimal (j_q, θ_q) for Maximizing $\Delta H(q, \mathbf{D}^{(q)})$

IF $x[j_q] \geq \theta_q$ APPEND $x \rightarrow \mathbf{D}_R^{(q)}$ ELSE APPEND $x \rightarrow \mathbf{D}_L^{(q)}$

INPUT: $\mathbf{D}_L^{(q)} \rightarrow q_L$ AND $\mathbf{D}_R^{(q)} \rightarrow q_R$

Splitting is Performed Recursively until Termination at All Leaf Nodes

The Dataset

6	-4	4	7	0	-7	1	-8	0	-7
8	-9	-4	-5	0	8	10	-1	6	7
+1	+1	+1	+1	+1	-1	-1	-1	-1	-1

Decision Tree: Learning C4.5 Splits

Sorted in Ascending Order along $d = 1$	6	-4	4	7	0	-7	1	-8	0	-7
	8	-9	-4	-5	0	8	10	-1	6	7
	+1	+1	+1	+1	+1	-1	-1	-1	-1	-1
	-8	-7	-7	-4	0	0	1	4	6	7
	-1	7	8	-9	0	6	10	-4	8	-5
	-1	-1	-1	+1	+1	-1	-1	+1	+1	+1

$$X[j] \geq \theta$$

Estimating Impurity Drop

$j = 1$

$\theta = 0$

-8	-7	-7	-4	0	0	1	4	6	7
-1	7	8	-9	0	6	10	-4	8	-5
-1	-1	-1	+1	+1	-1	-1	+1	+1	+1

Left Split
Right Split

$$H_{LR}(j, \theta) = P_L \text{Im}(D_L) + P_R \text{Im}(D_R)$$

$$H_{LR}(j = 1, \theta = 0) = \frac{4}{10} \times \frac{1}{4} + \frac{6}{10} \times \frac{2}{6} = 0.3$$

Decision Tree: Learning C4.5 Splits

$x[j = 1] \geq \theta$	-8	-7	-7	-4	0	0	1	4	6	7
	-1	7	8	-9	0	6	10	-4	8	-5
	-1	-1	-1	+1	+1	-1	-1	+1	+1	+1

θ	$P_L \text{Im}(D_L) + P_R \text{Im}(D_R)$	H_c
-7	$\frac{1}{10} \times 0 + \frac{9}{10} \times \frac{4}{9}$	$\frac{4}{10}$

Decision Tree: Learning C4.5 Splits

$x[j = 1] \geq \theta$	-8	-7	-7	-4	0	0	1	4	6	7
	-1	7	8	-9	0	6	10	-4	8	-5
	-1	-1	-1	+1	+1	-1	-1	+1	+1	+1

θ	$P_L \text{Im}(D_L) + P_R \text{Im}(D_R)$	H_c
-7	$\frac{1}{10} \times 0 + \frac{9}{10} \times \frac{4}{9}$	$\frac{4}{10}$
-4	$\frac{3}{10} \times 0 + \frac{7}{10} \times \frac{2}{7}$	$\frac{2}{10}$

Decision Tree: Learning C4.5 Splits

$x[j = 1] \geq \theta$	-8	-7	-7	-4	0	0	1	4	6	7
	-1	7	8	-9	0	6	10	-4	8	-5
	-1	-1	-1	+1	+1	-1	-1	+1	+1	+1

θ	$P_L \text{Im}(D_L) + P_R \text{Im}(D_R)$	H_c
-7	$\frac{1}{10} \times 0 + \frac{9}{10} \times \frac{4}{9}$	$\frac{4}{10}$
-4	$\frac{3}{10} \times 0 + \frac{7}{10} \times \frac{2}{7}$	$\frac{2}{10}$
0	$\frac{4}{10} \times \frac{1}{4} + \frac{6}{10} \times \frac{2}{6}$	$\frac{3}{10}$

Decision Tree: Learning C4.5 Splits

$x[j = 1] \geq \theta$	-8	-7	-7	-4	0	0	1	4	6	7
	-1	7	8	-9	0	6	10	-4	8	-5
	-1	-1	-1	+1	+1	-1	-1	+1	+1	+1

θ	$P_L \text{Im}(D_L) + P_R \text{Im}(D_R)$	H_c
-7	$\frac{1}{10} \times 0 + \frac{9}{10} \times \frac{4}{9}$	$\frac{4}{10}$
-4	$\frac{3}{10} \times 0 + \frac{7}{10} \times \frac{2}{7}$	$\frac{2}{10}$
0	$\frac{4}{10} \times \frac{1}{4} + \frac{6}{10} \times \frac{2}{6}$	$\frac{3}{10}$
1	$\frac{6}{10} \times \frac{2}{6} + \frac{4}{10} \times \frac{1}{4}$	$\frac{3}{10}$

Decision Tree: Learning C4.5 Splits

$x[j = 1] \geq \theta$

-8	-7	-7	-4	0	0	1	4	6	7
-1	7	8	-9	0	6	10	-4	8	-5
-1	-1	-1	+1	+1	-1	-1	+1	+1	+1

θ	$P_L Im(D_L) + P_R Im(D_R)$	H_c
-7	$\frac{1}{10} \times 0 + \frac{9}{10} \times \frac{4}{9}$	$\frac{4}{10}$
-4	$\frac{3}{10} \times 0 + \frac{7}{10} \times \frac{2}{7}$	$\frac{2}{10}$
0	$\frac{4}{10} \times \frac{1}{4} + \frac{6}{10} \times \frac{2}{6}$	$\frac{3}{10}$
1	$\frac{6}{10} \times \frac{2}{6} + \frac{4}{10} \times \frac{1}{4}$	$\frac{3}{10}$
4	$\frac{7}{10} \times \frac{2}{7} + \frac{3}{10} \times 0$	$\frac{2}{10}$

Decision Tree: Learning C4.5 Splits

$x[j = 1] \geq \theta$

-8	-7	-7	-4	0	0	1	4	6	7
-1	7	8	-9	0	6	10	-4	8	-5
-1	-1	-1	+1	+1	-1	-1	+1	+1	+1

θ	$P_L Im(D_L) + P_R Im(D_R)$	H_c
-7	$\frac{1}{10} \times 0 + \frac{9}{10} \times \frac{4}{9}$	$\frac{4}{10}$
-4	$\frac{3}{10} \times 0 + \frac{7}{10} \times \frac{2}{7}$	$\frac{2}{10}$
0	$\frac{4}{10} \times \frac{1}{4} + \frac{6}{10} \times \frac{2}{6}$	$\frac{3}{10}$
1	$\frac{6}{10} \times \frac{2}{6} + \frac{4}{10} \times \frac{1}{4}$	$\frac{3}{10}$
4	$\frac{7}{10} \times \frac{2}{7} + \frac{3}{10} \times 0$	$\frac{2}{10}$
6	$\frac{8}{10} \times \frac{3}{8} + \frac{2}{10} \times 0$	$\frac{3}{10}$

Decision Tree: Learning C4.5 Splits

Sorted in Ascending Order along $d = 2$

-4	7	4	-8	0	0	-7	-7	6	1
-9	-5	-4	-1	0	6	7	8	8	10
+1	+1	+1	-1	+1	-1	-1	-1	+1	-1

$$x[j = 2] \geq \theta$$

θ	-5	-4	-1	0	6	7	8
H_c	$\frac{4}{10}$	$\frac{3}{10}$	$\frac{2}{10}$	$\frac{3}{10}$	$\frac{2}{10}$	$\frac{3}{10}$	$\frac{4}{10}$

Decision Tree: Learning C4.5 Splits

$d = 1$	6	-4	4	7	0	-7	1	-8	0	-7	$\sigma = 5.5136$
$d = 2$	8	-9	-4	-5	0	8	10	-1	6	7	$\sigma = 6.6332$
	+1	+1	+1	+1	+1	-1	-1	-1	-1	-1	

$median[d = 2] = 3$

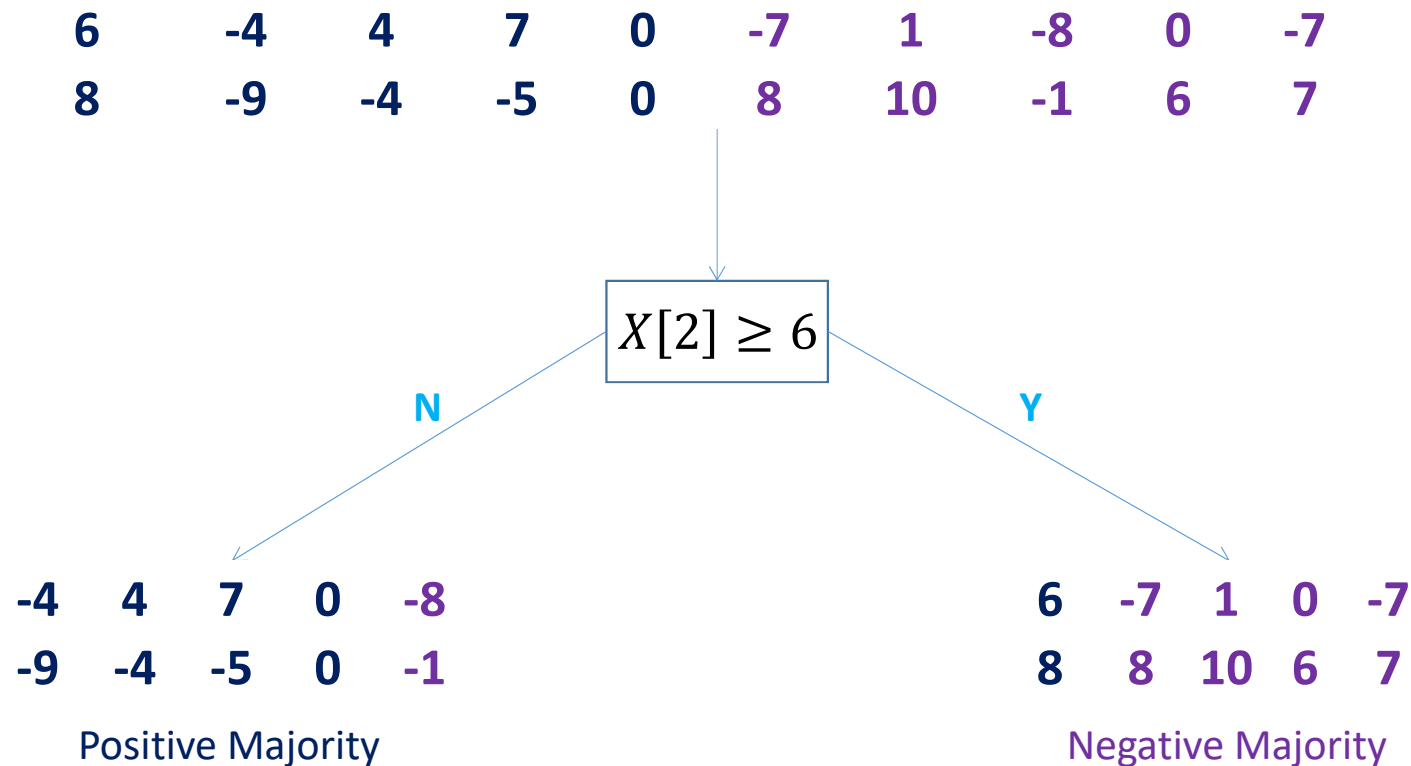
$$x[j = 1] \geq \theta$$

θ	-7	-4	0	1	4	6
H_c	$\frac{4}{10}$	$\frac{2}{10}$	$\frac{3}{10}$	$\frac{3}{10}$	$\frac{2}{10}$	$\frac{3}{10}$

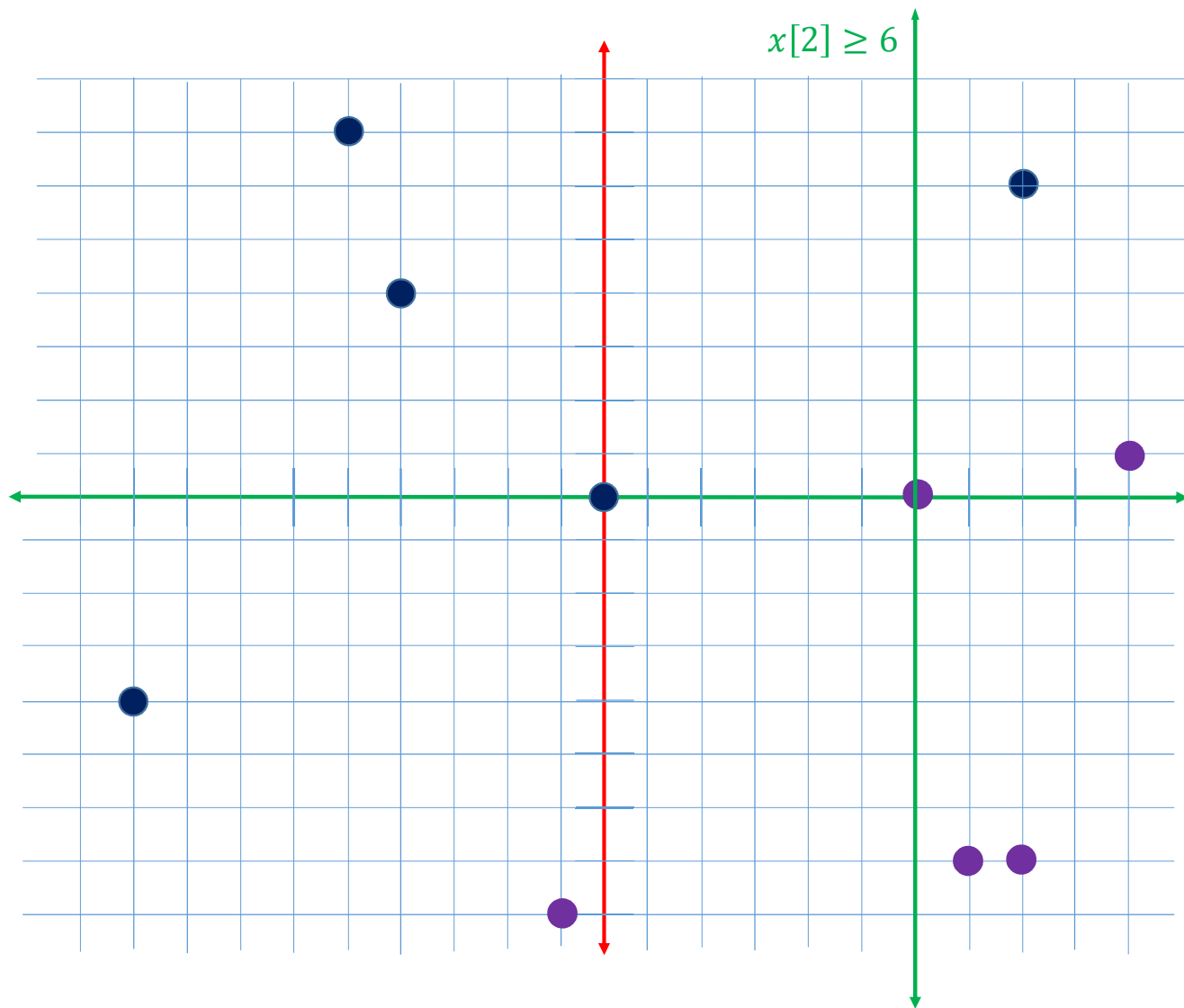
$$x[j = 2] \geq \theta$$

θ	-5	-4	-1	0	6	7	8
H_c	$\frac{4}{10}$	$\frac{3}{10}$	$\frac{2}{10}$	$\frac{3}{10}$	$\frac{2}{10}$	$\frac{3}{10}$	$\frac{4}{10}$

Decision Tree: Split at Root Node

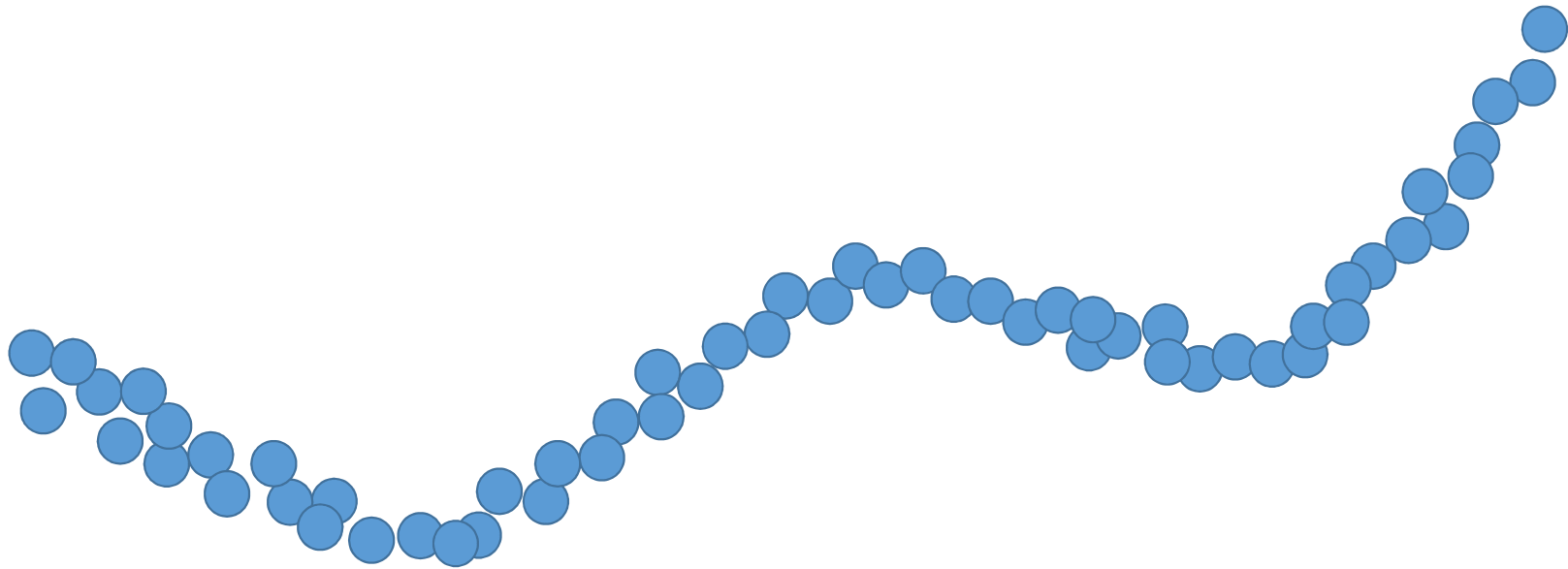


Assignment: Create Split Nodes for Next Depth and Classify $x = \begin{bmatrix} -10 \\ 10 \end{bmatrix}$

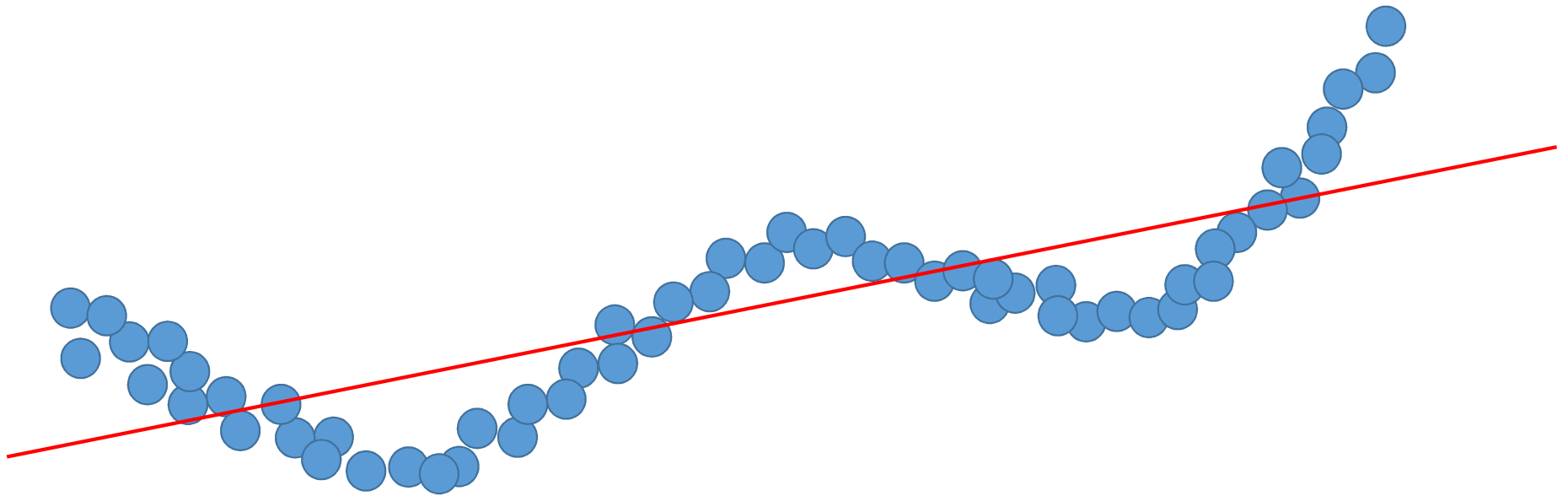


6	-4	4	7	0	-7	1	-8	0	-7
8	-9	-4	-5	0	8	10	-1	6	7
+1	+1	+1	+1	+1	-1	-1	-1	-1	-1

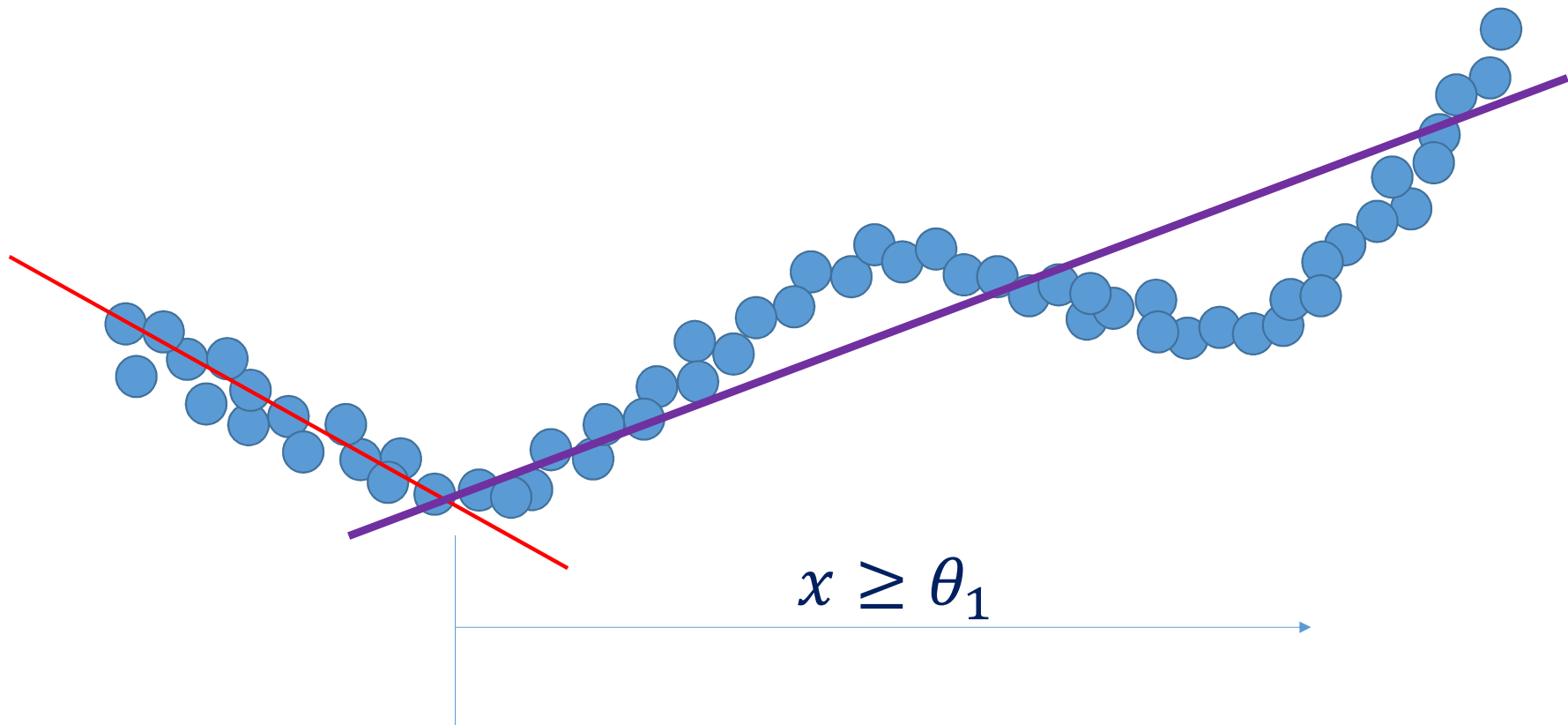
Piece-wise Linear Regression



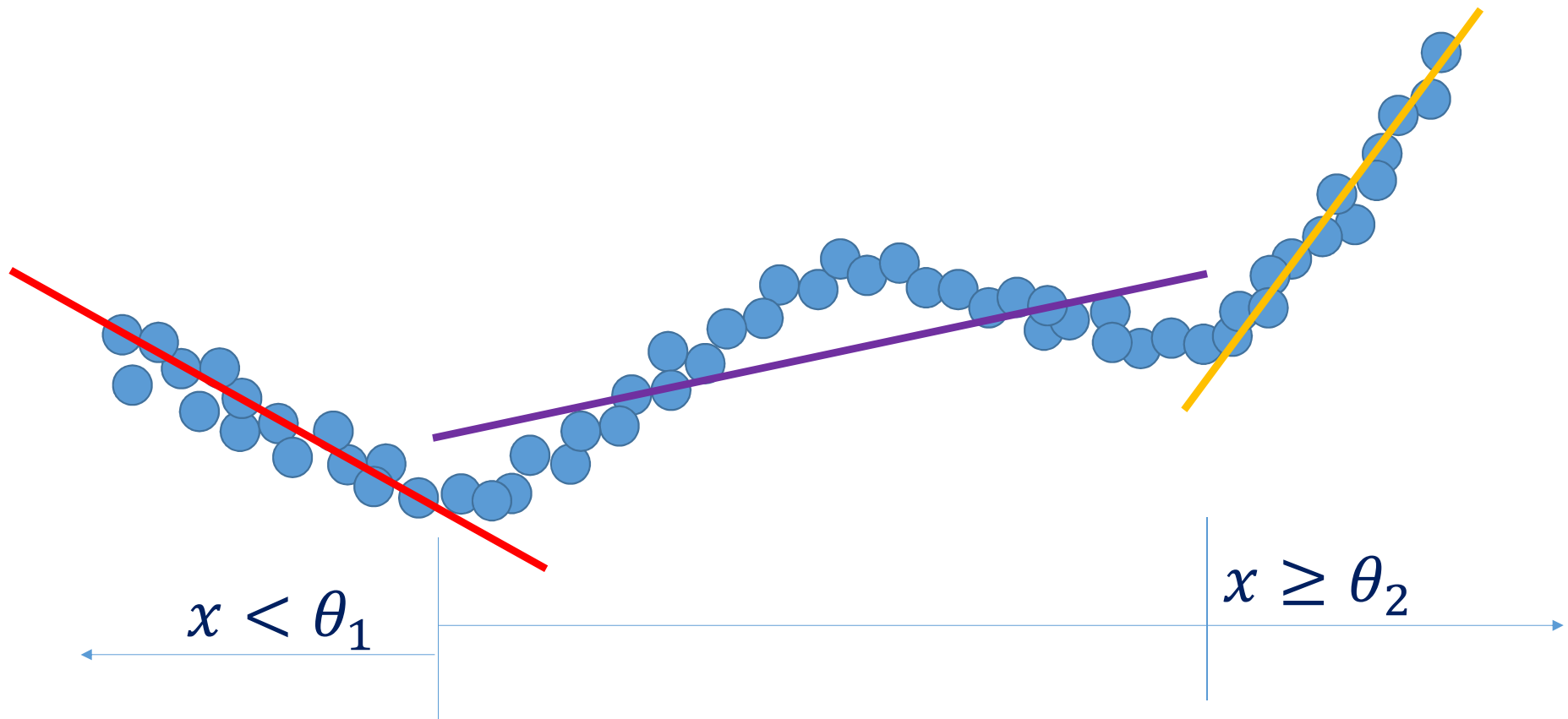
Piece-wise Linear Regression



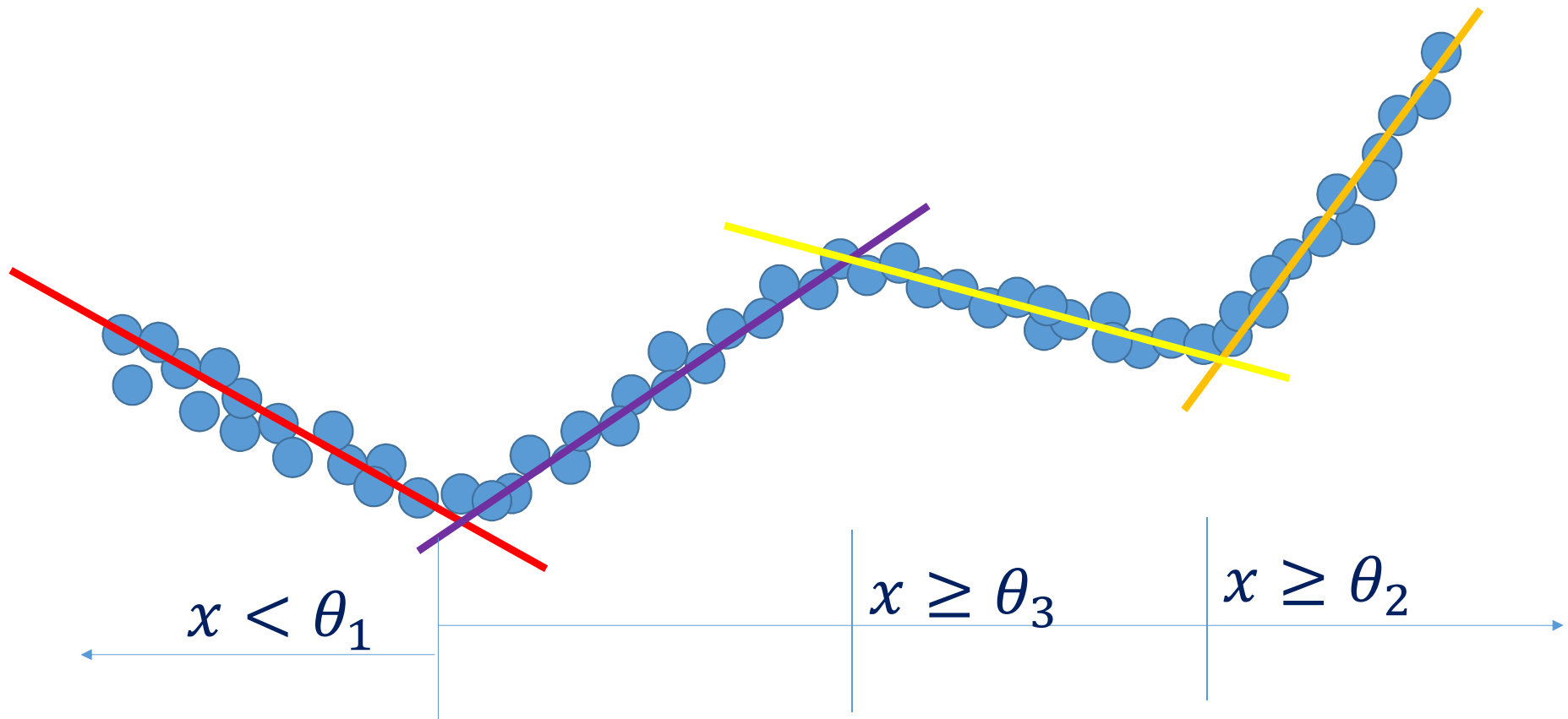
Piece-wise Linear Regression



Piece-wise Linear Regression

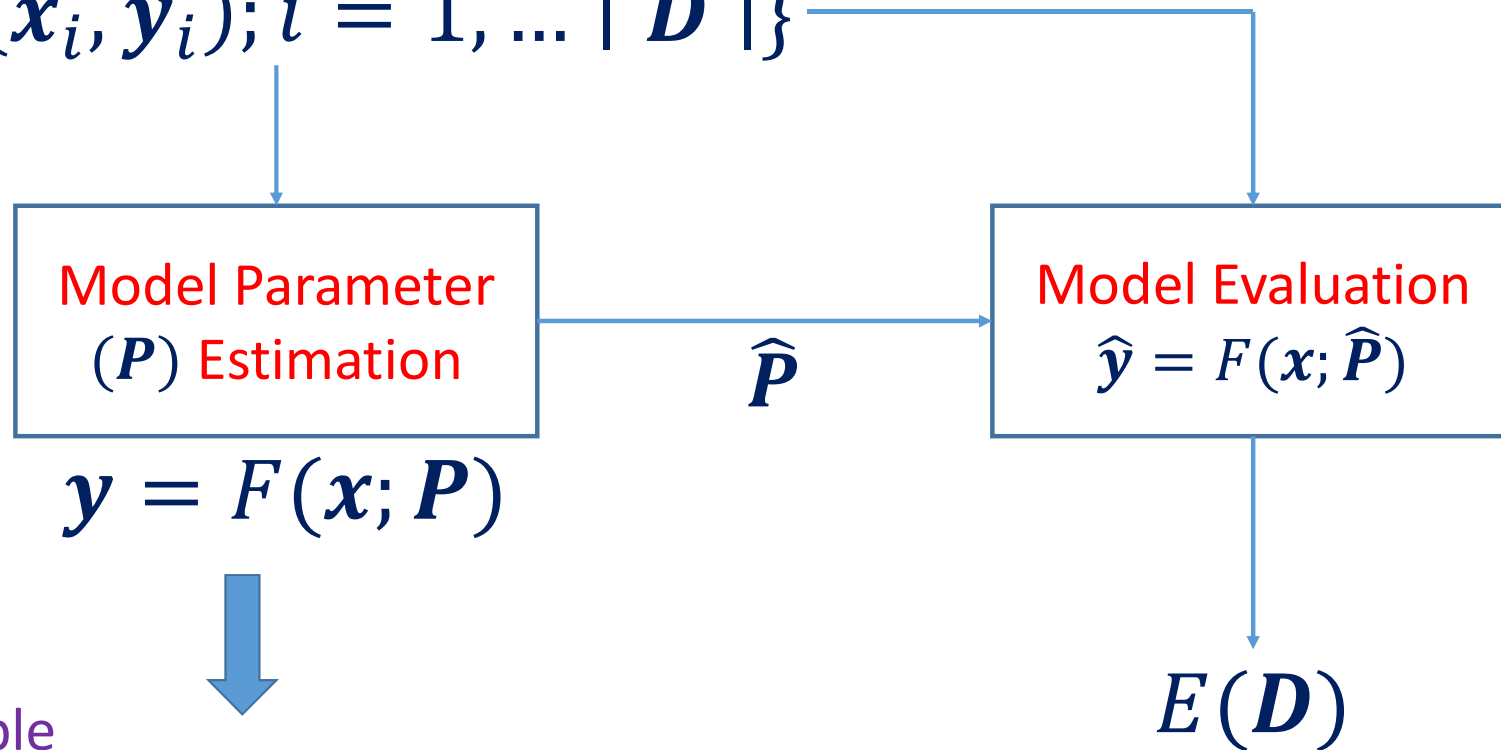


Piece-wise Linear Regression



Regression Model

$$\mathbf{D} = \{(\mathbf{x}_i, \mathbf{y}_i); i = 1, \dots | \mathbf{D} | \}$$

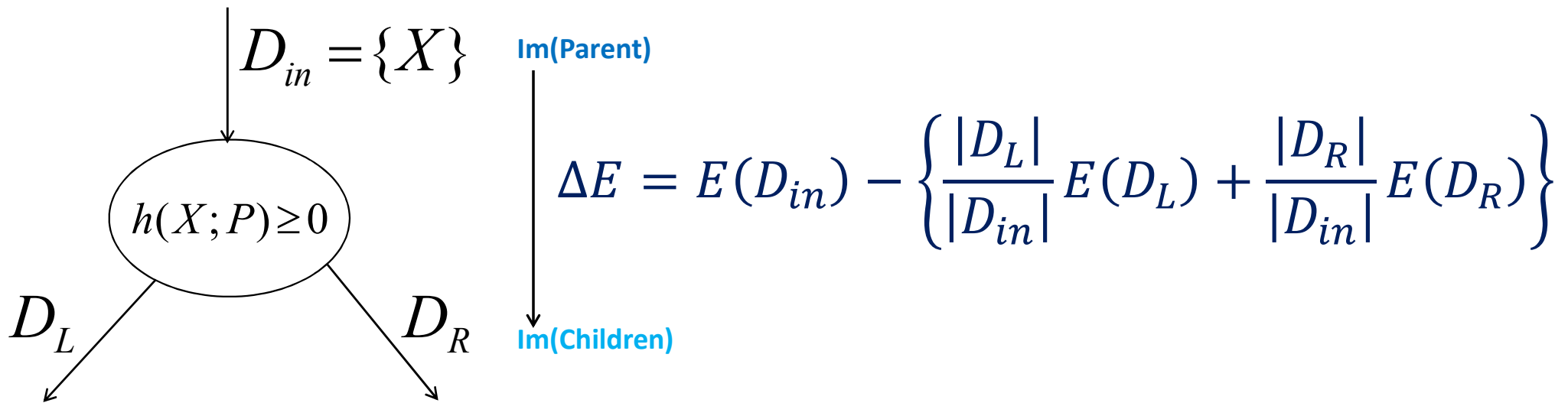


Example

$$\mathbf{y} = \mathbf{A}\mathbf{x} + \mathbf{b} : \mathbf{P} = \{\mathbf{A}, \mathbf{b}\}$$

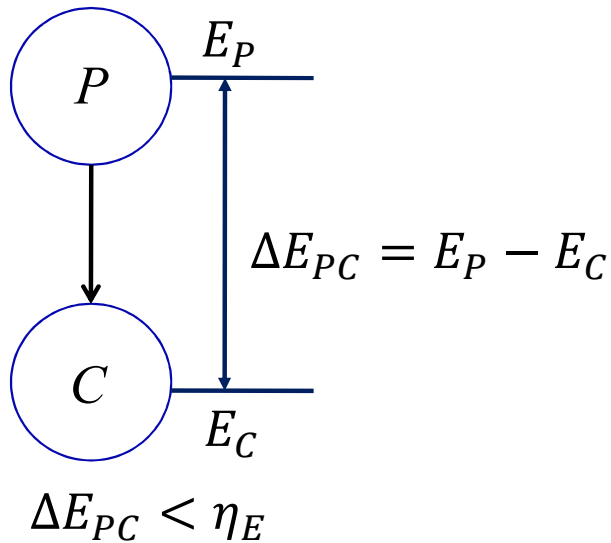
Regression Tree: Error

$$D_{in} = \{(\mathbf{x}_i, \mathbf{y}_i); i = 1, \dots | D_{in} |\}$$

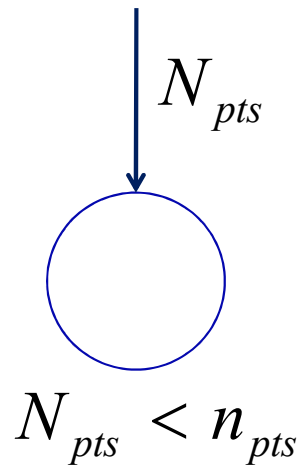


Split Objective: Maximize Error Drop

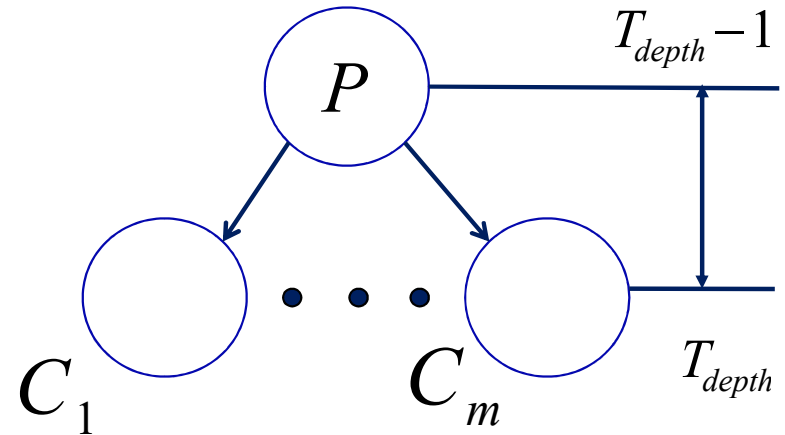
Termination Criteria



**Insufficient Error Drop
from Parent to Child Node**



**Insufficient Input
to Node**



**Nodes at Maximum Allowable
Tree Depth**

Regression Tree Construction

1. Input to Node q is $D^{(q)}$
2. IF $checkTerminationConditions(q) = TRUE$ THEN Tag q as Leaf Node
3. IF q is Leaf Node THEN Estimate Model Parameters $\hat{P}_q$ using $D^{(q)}$
4. ELSE Search for Optimal (j_q, θ_q) for Maximizing $\Delta E(q, D^{(q)})$
5. IF $x[j_q] \geq \theta_q$ PUSH $x \rightarrow D_R^{(q)}$; ELSE PUSH $x \rightarrow D_L^{(q)}$
6. Split $D^{(q)}$ into $D_L^{(q)}$ & $D_R^{(q)}$: $D^{(q)} = D_L^{(q)} \cup D_R^{(q)}$ & $D_L^{(q)} \cap D_R^{(q)} = \varphi$
7. Children Nodes of q are q_L & q_R with Respective Inputs $D_L^{(q)}$ & $D_R^{(q)}$
8. Splitting is Performed Recursively until Termination at All Leaf Nodes

Summary

- Top-Down Hierarchical Feature Space Partitioning
- Classification Trees: Impurity Drop Maximization
- Three Impurity Measures
- Axis Aligned Decision Functions at Split Nodes
- Dominant Class Labels at Leaf Nodes
- Regression Trees: Error Drop Maximization
- Axis Aligned Decision Functions at Split Nodes
- Learned Models at Leaf Nodes



Thank You